MIL-A-27670C(USAF)
16 August 1965
Superseding
MIL-A-27670B(USAF)
1 March 1963

MILITARY SPECIFICATION

AMPLIFIER-INDICATOR GROUP, INDICATED AIRSPEED A/A24G-10

1. SCOPE

1.1 This specification covers the requirements for one type of aircraft indicated airspeed amplifier-indicator group, designated A/A24G-10.

*2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-25627	Transmitter, Synchro, Angle of Attack or Side Slip
MIL-M-26512	Maintainability Program Requirements for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-C-38037	Computer, Central Air Data CPU-43/A
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

FSC 6610

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Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS17322	Meter, Time Totalizing Miniature, 400 Cycles, Digital
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Aeronautical Radio, Incorporated

Standard ARINC 545 Subsonic Air Data Computer System

(Copies of the ARINC document may be obtained from Aeronautical Radio, Incorporated, 1700 K Street N. W., Washington 6, D. C)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The amplifier-indicator shall consist of:

Indicator, Airspeed Mach Safe Speed AVK-16/A24G-10
Amplifier, Electronic Control ASK-10/A24G-10

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. If used within a hermetically sealed inclosure, fungicidal treatment shall not be required.

3.4.2 Metals. Metals shall be of a corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

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3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the instrument, except where magnetic materials are essential.

3.4.4 Protective treatment. Materials that are utilized within the hermetically sealed case are suitably protected against corrosion.

3.4.5 Toxic and corrosive fumes. The materials, as installed and under the service conditions specified, shall not liberate gases or fumes which will result in corrosion that causes malfunction of equipment or discoloration of dials or indicia. Toxic gases or fumes that are detrimental to performance of the aircraft or health of personnel shall not be liberated under the service conditions specified herein.

3.5 Design and construction. The amplifier-indicator group shall be designed and constructed in accordance with figures 1 through 14 and shall have reliability consistent with the specified requirements and minimum weight specified herein.

3.5.1 Life. The equipment shall be designed to have a reliable life of at least 1,000 hours without requiring servicing. The equipment shall be designed to have a minimum operating life of at least 2,000 hours with reasonable servicing and replacement of parts. The contractor shall prepare a list of parts requiring replacement within the minimum operating life of the equipment. Life requirements of lamps shall be in accordance with applicable specifications approved by the procuring activity.

3.5.2 Presentation

3.5.2.1 Method of presentation. The method of presenting the information to the pilot shall be by means of a moving scale to be read against a fixed-center reference line for the aircraft mach number, indicated airspeed, and safe speed. The command and limit indications shall be by means of translational motion of the symbols along the graduated vertical scales. The scales and symbols shall be arranged as shown on figure 1.

3.5.2.2 The indicator shall present the following information derived from signal inputs in the form of electrical synchro signals from a central air data computer:

a. Aircraft mach number: Range, 0.2 to 1.0

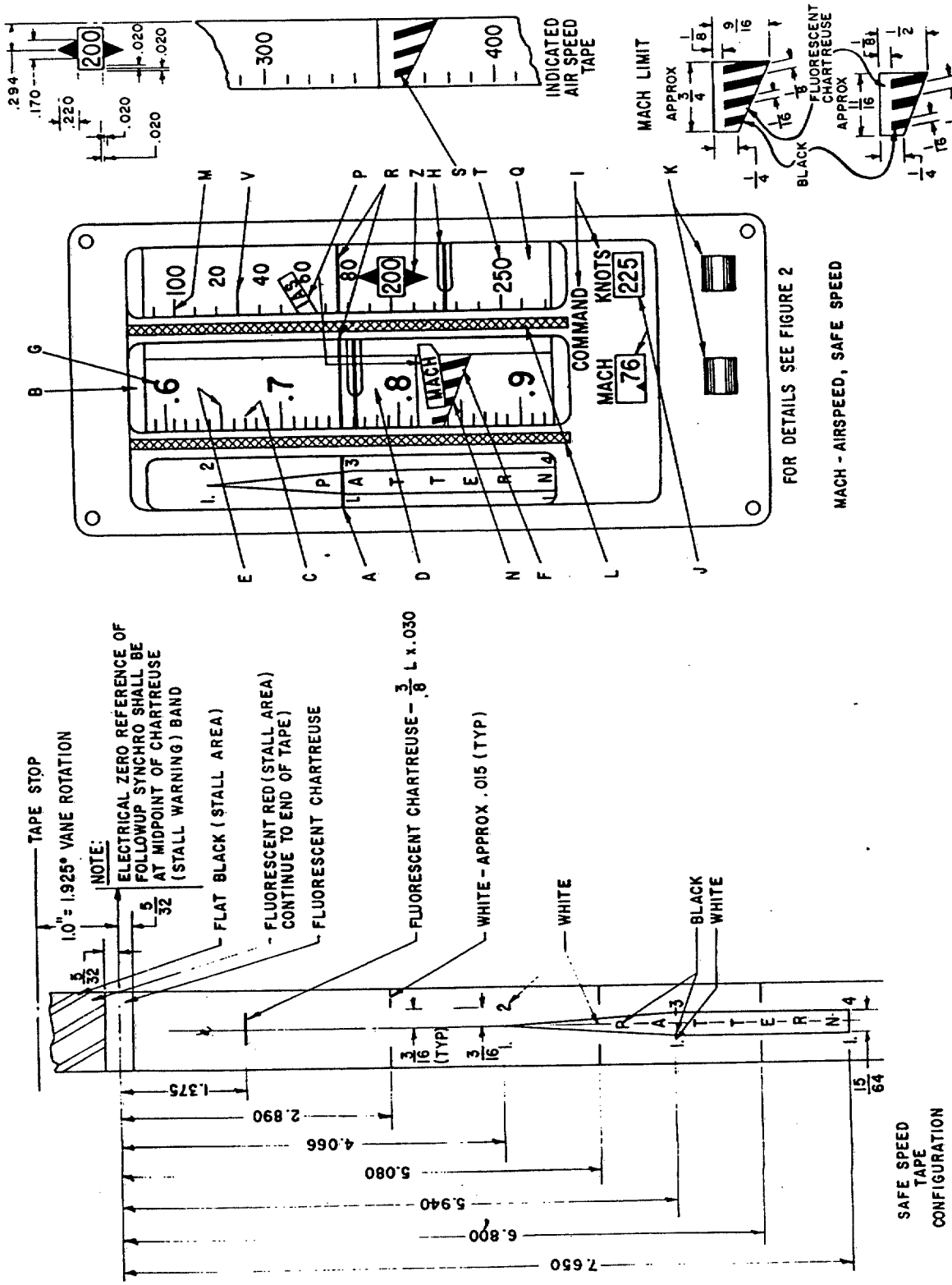
b. Indicated airspeed: Range 50 to 600 knots

c. Safe speed: (In terms of angle of attack) -10° to +25° from an angle of attack transmitter

d. Command mach number: Range 0.2 to 1.0

e. Command airspeed: Range 50-600 knots.

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★ FIGURE 1. Presentation

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A 0.030
B Black
C 3/16 White
D Black tape
E 5/16 White
F Black
G 5/16 White numerals
H White command marker with 0.030 black centerline total width 1/8
This marker shall be located below the fixed read reference line
I 1/8 White lettering to be illuminated by instrument lighting
J 1/4 White numerals
K Spring centered switch gray
L Gray
M 3/16 White
N Chartreuse fluorescent
P Red 3/16 x 9/16 black letters 1/8 high; location optional within 3/4 lubber line
Q Black tape
R 0.030 White fixed read reference
S IAS limit indication: Chartreuse and black diagonal lines
T 7/32 White numerals
V 1/8 White
Y Mach limit indication: Chartreuse and black diagonal lines
Z White

Letters and numerals shall conform to MS33558. Stroke width of numerals and grads 1/16 to 1/8 of height; width of numerals to be 3/5 of height

All colors to conform to FED-STD-595

Black, color No. 37038, White, color No. 37875, Gray, color No. 36231

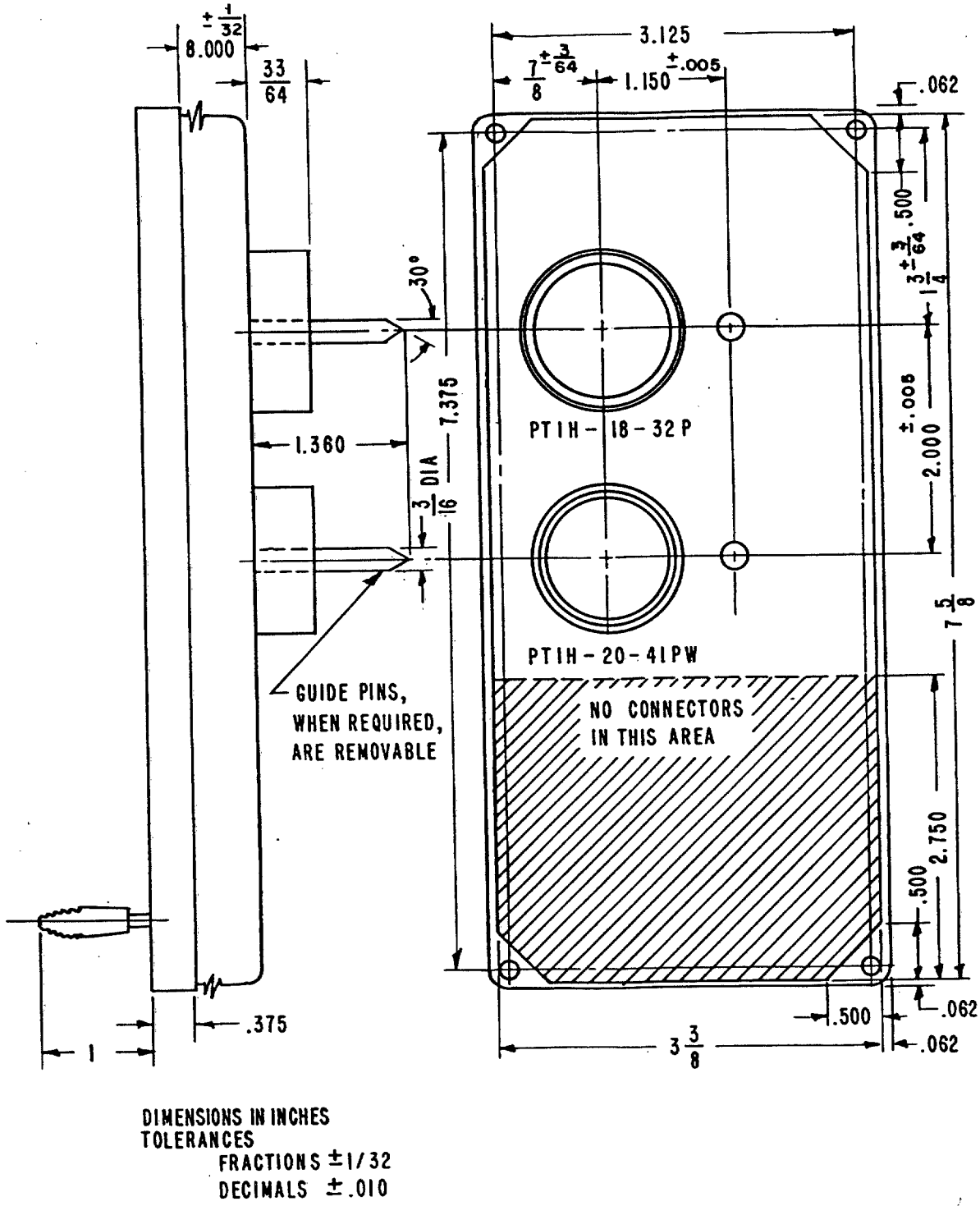
Red - Red Fluorescent 116-1548 (Switzer Brothers, Cleveland, Ohio or equal)

Chartreuse - Chartreuse Fluorescent (Switzer Brothers Saturn Yellow, or equal)

DIMENSIONS IN INCHES

* FIGURE 2. Dial And Case Details

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* FIGURE 3. Case, Mach-Airspeed Rear & Side View

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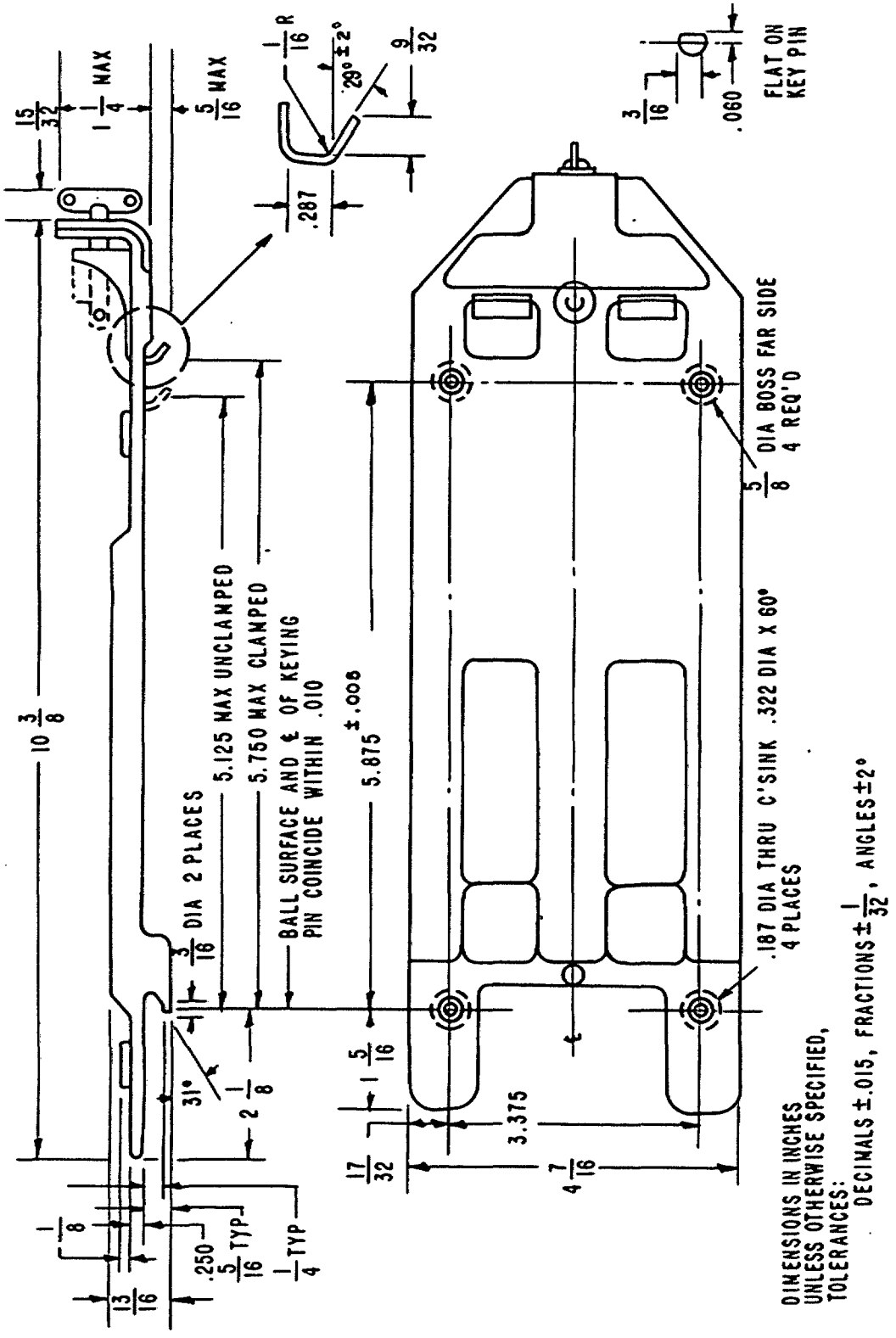


FIGURE 4. Mounting Base.

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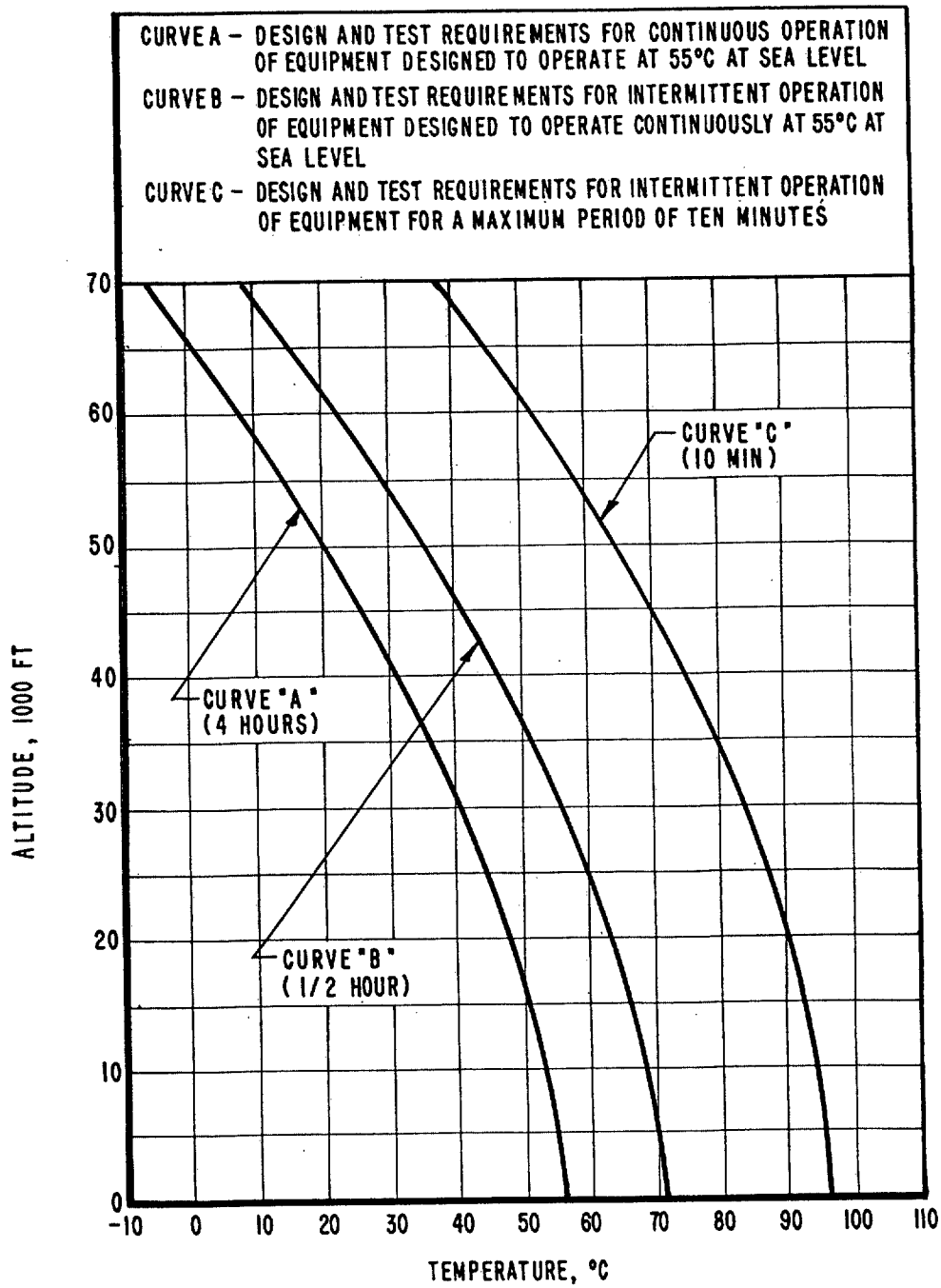


FIGURE 5. Curves For Altitude VS Temperature

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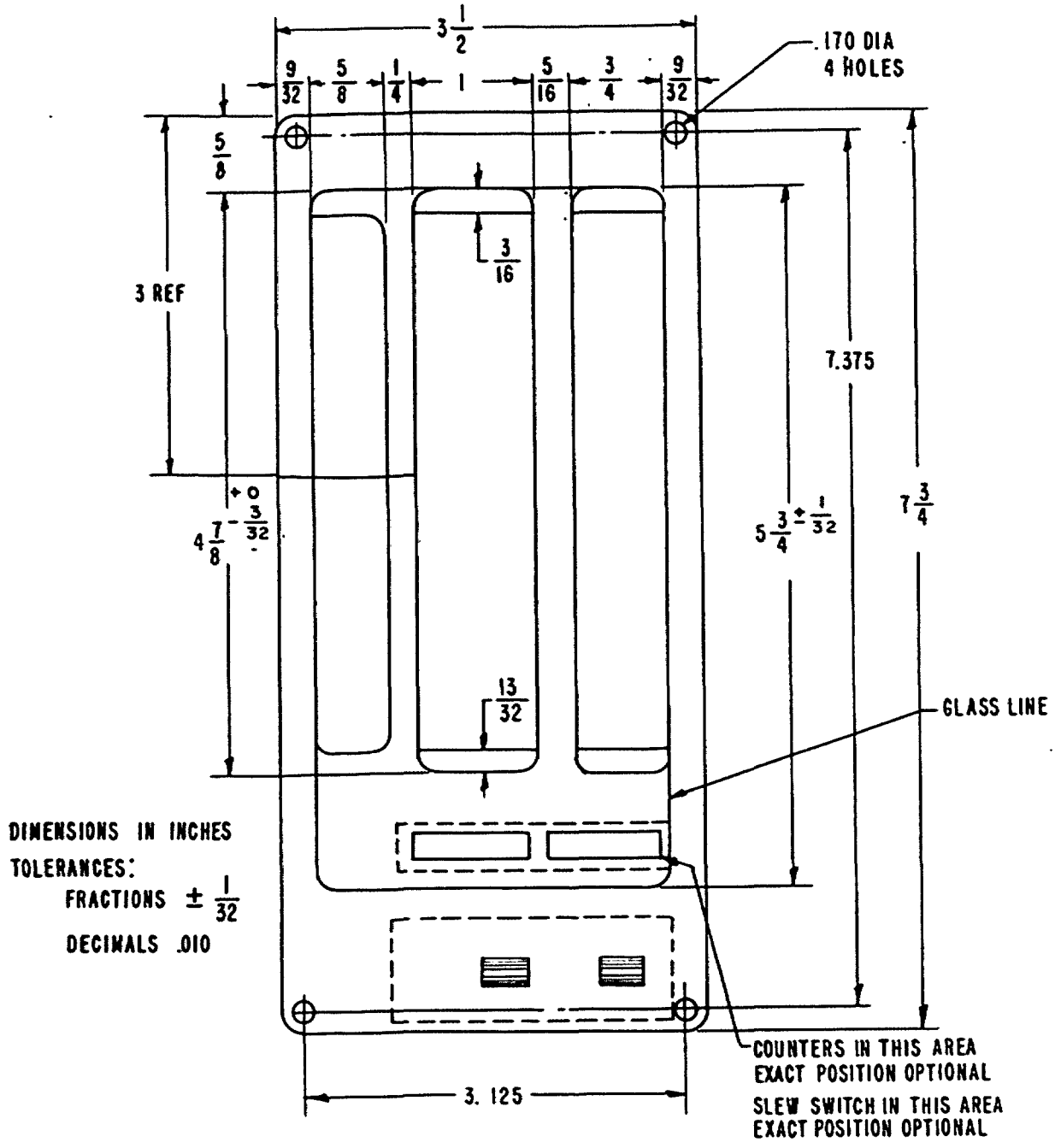


*FIGURE 6. Bezel, Mach-Airspeed-Safe Speed

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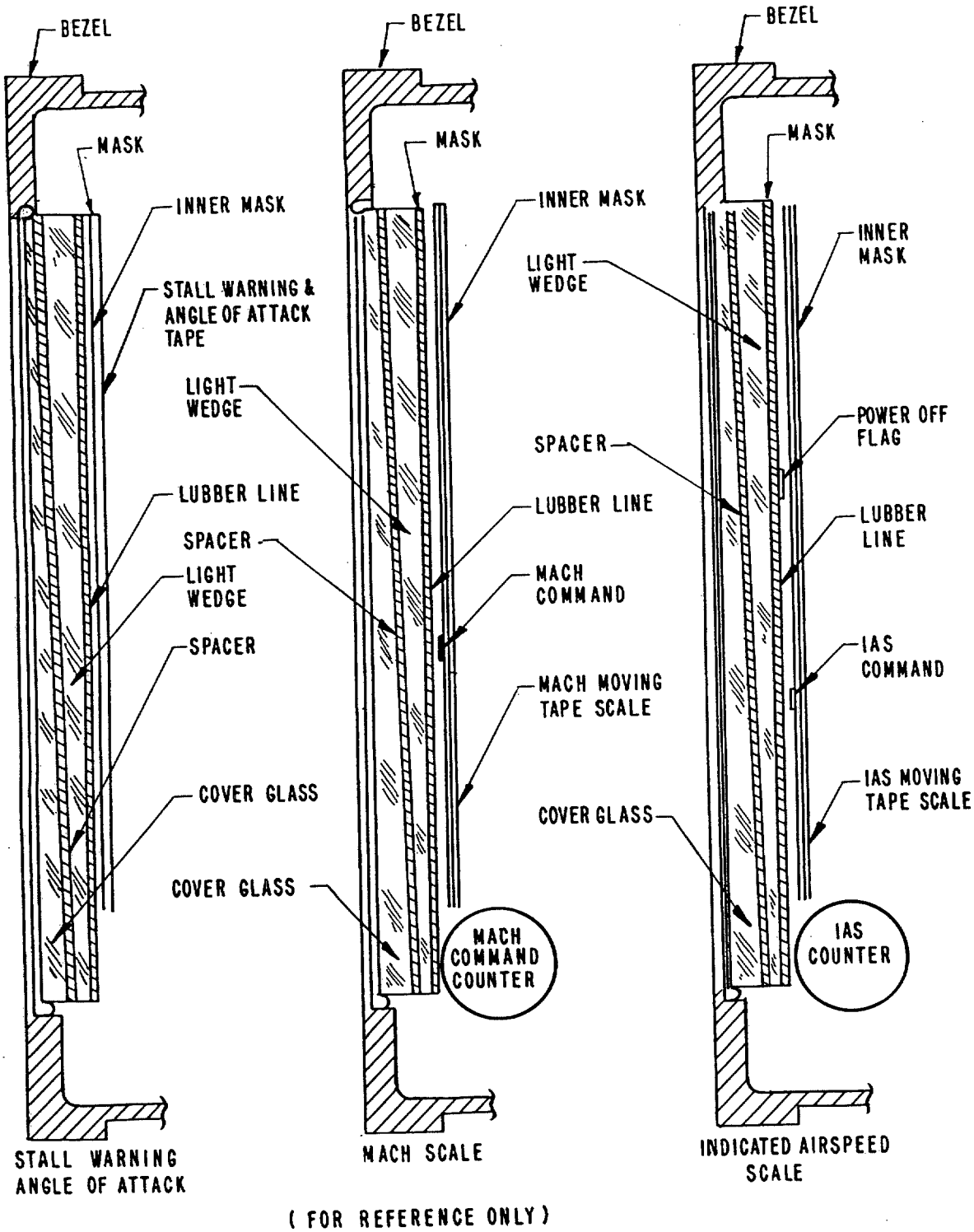


FIGURE 7. Sections, Mach Airspeed Safe Speed

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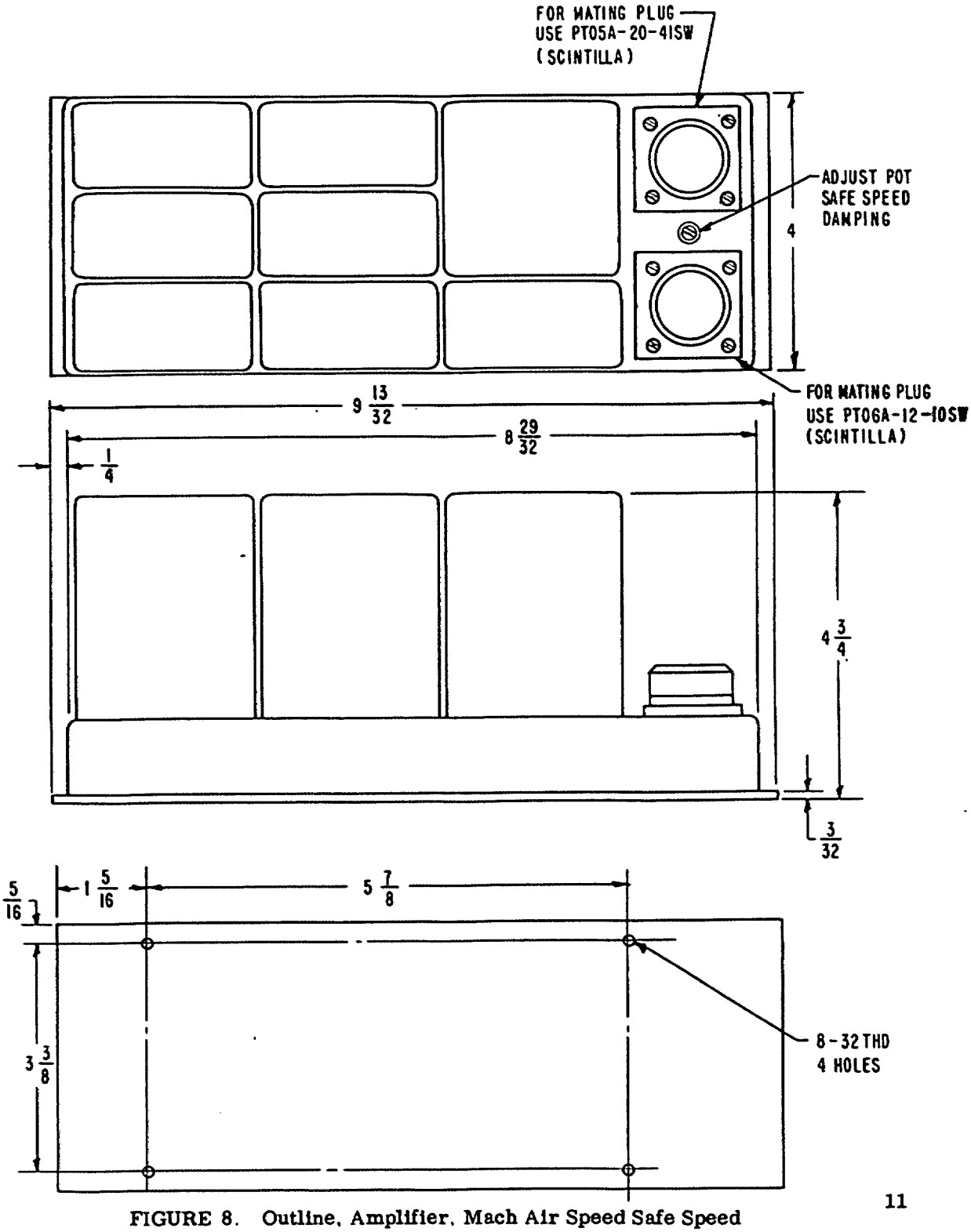


FIGURE 8. Outline, Amplifier, Mach Air Speed Safe Speed

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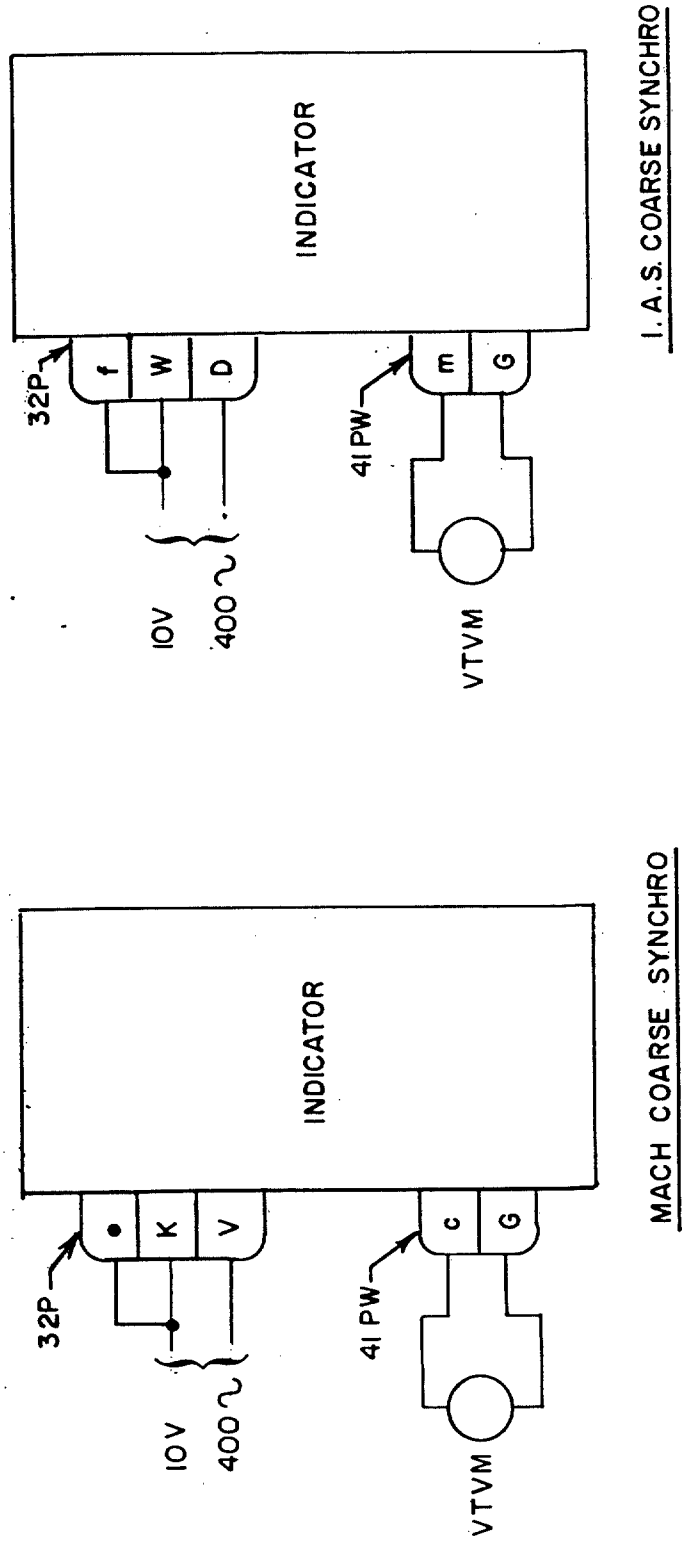
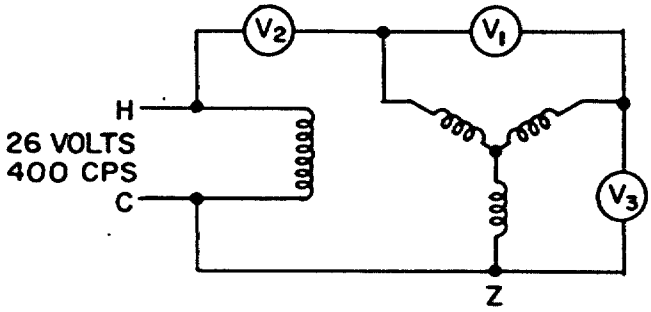


FIGURE 9. Synchro Correspondence

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V_1 — VACUUM TUBE VOLTMETER

V_2 — 1000 OHMS / VOLT VOLTMETER

V_3 — 1000 OHMS / VOLT VOLTMETER

*** FIGURE 10. Electrical Zero Circuit**

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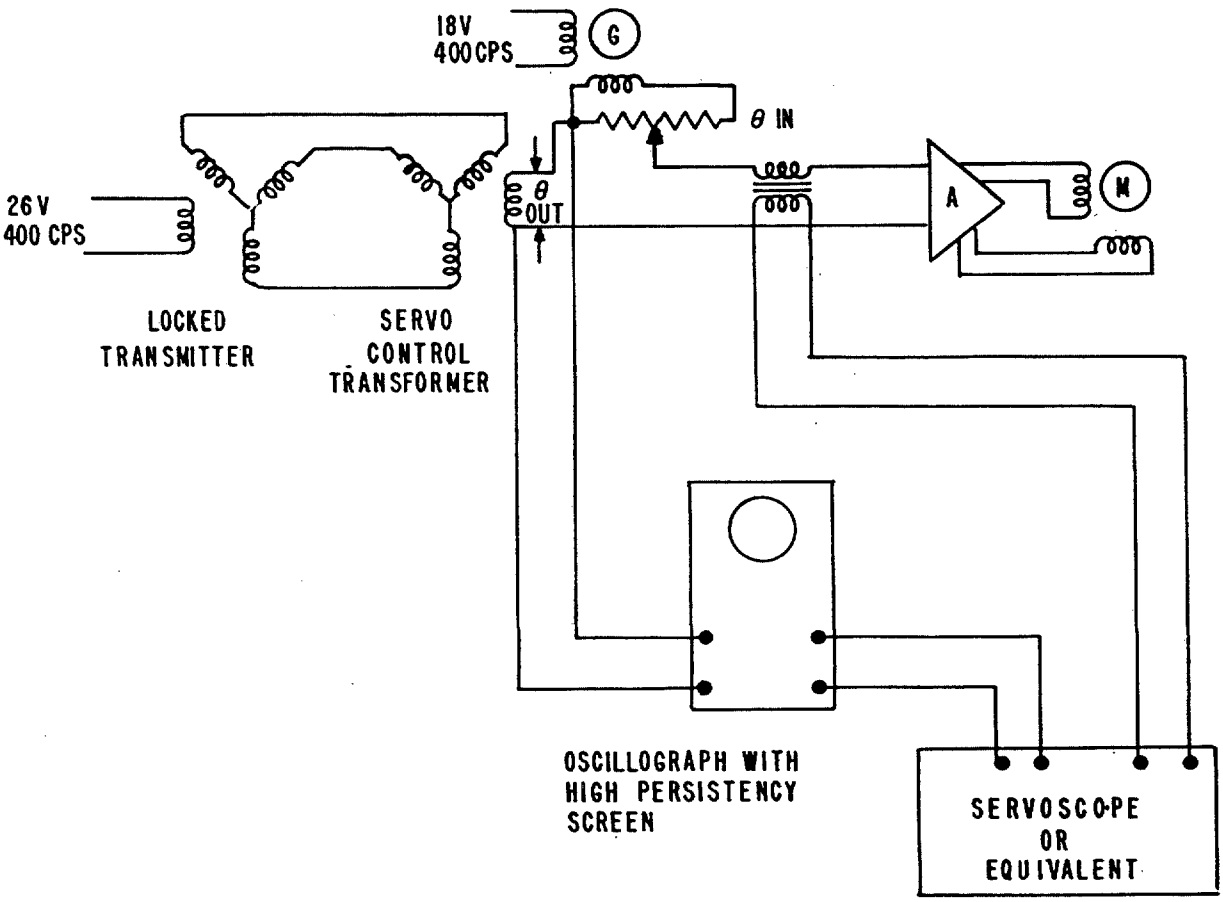


FIGURE 11. Frequency Response Circuit

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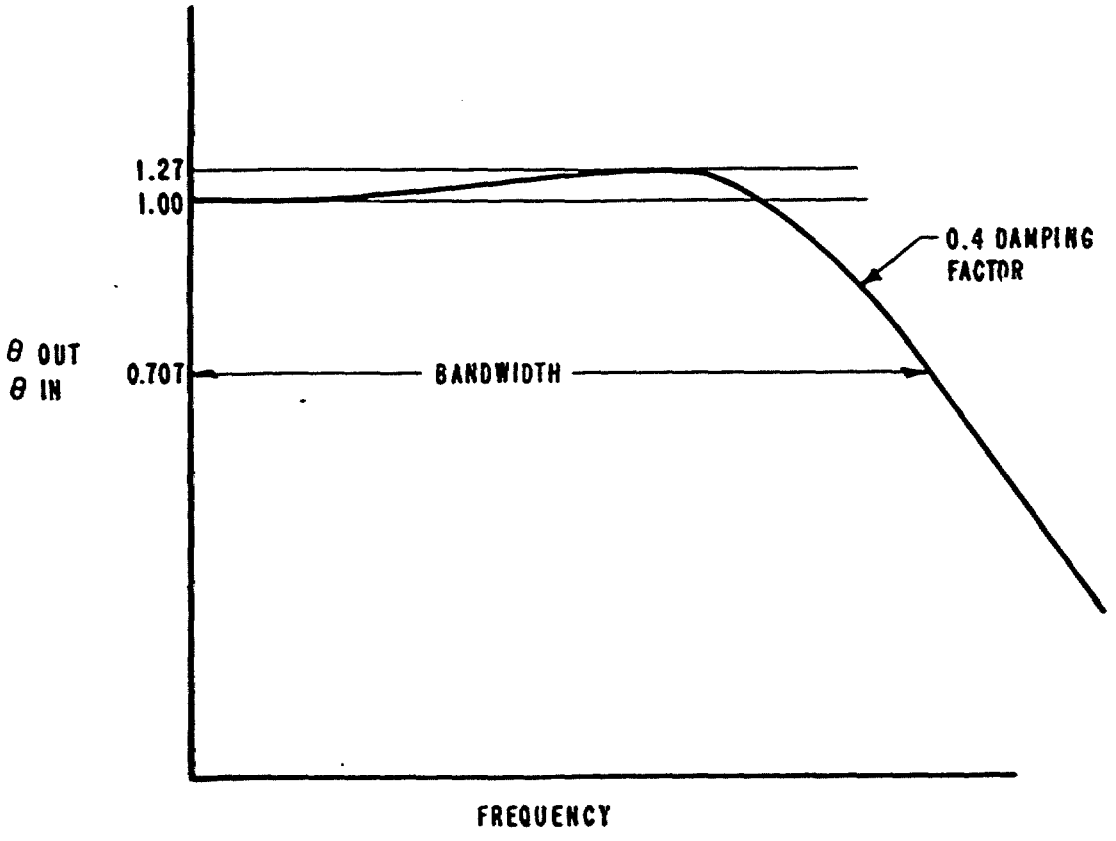


FIGURE 12. Band Width And Frequency

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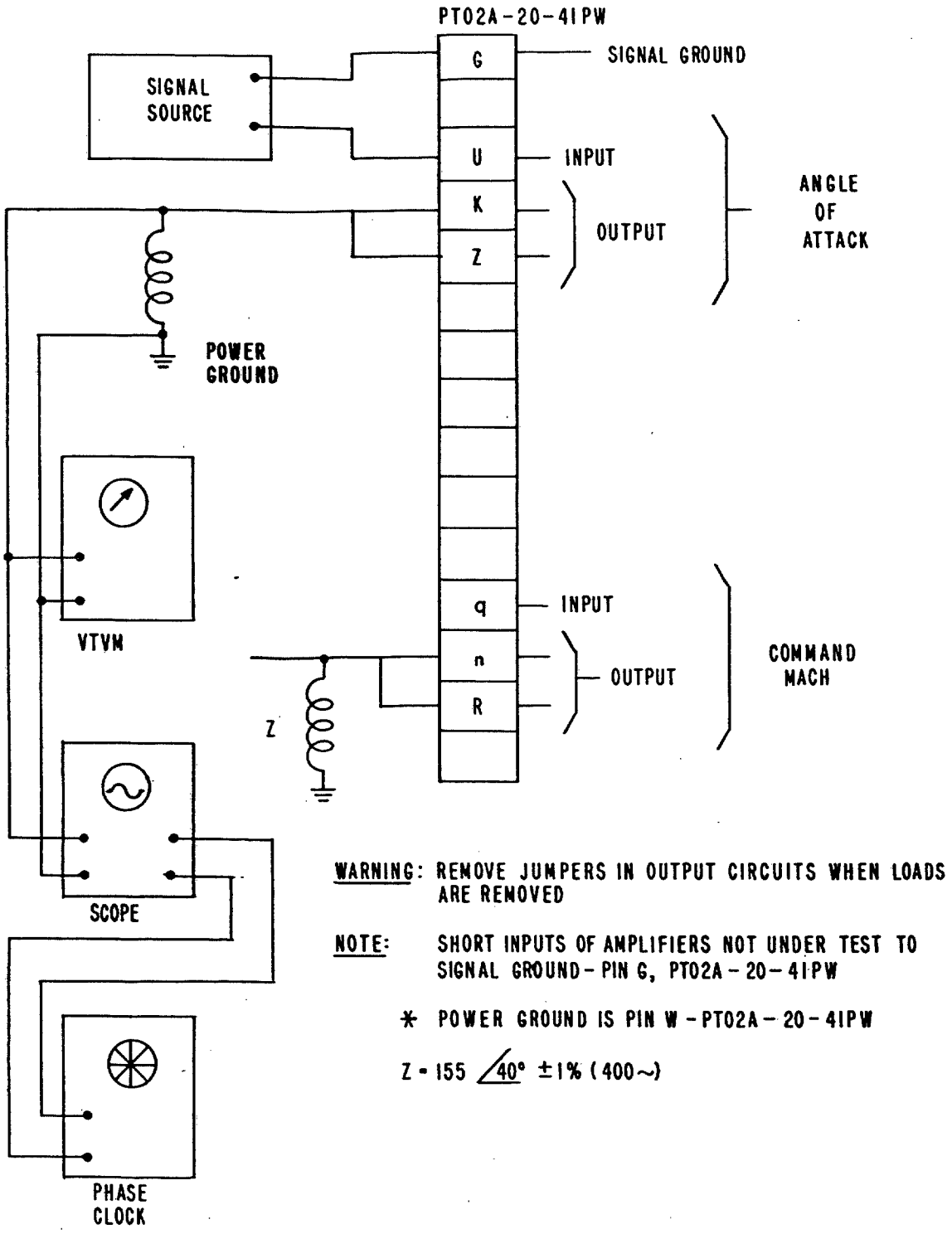


FIGURE 13. Circuit Test Connections

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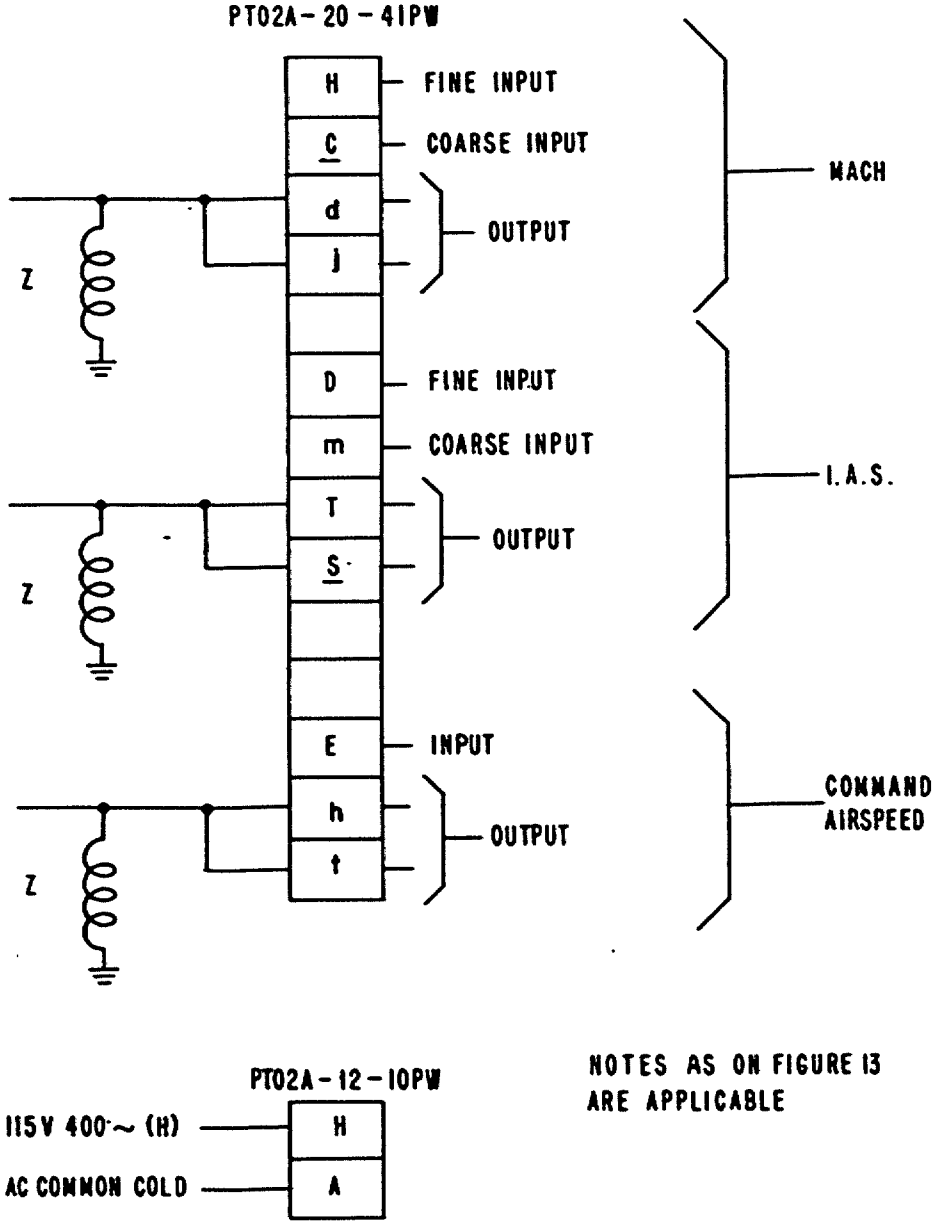


FIGURE 14. Circuit Test Connections

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* 3.5.2.2.1 Indicator synchros. Indicator synchros, or equivalent, shall be as follows:

a. Transmitting synchros (from CADC)

Aircraft mach - fine and coarse	AY50□42-A1
Indicated airspeed - fine and coarse	AY50□42-A1

b. Transmitting synchro (from angle of attack vane)

Angle of attack (safe speed) Clifton CGC-10-AS-7 or electrical equivalent

c. Indicator control transformer synchros (Receivers)

Aircraft mach	AY30□S-18-A1
Indicated airspeed	AY30□S-18-A1
Maximum safe speed	AY30□S-18-A1

3.5.3 Airspeed mach display. The airspeed-mach display shall conform to figure 1. The design shall be reasonably simple to permit overhaul or repair without requiring numerous special tools and fixtures. The amplifier-indicator group shall be so constructed as to withstand, without failure, the normal strains, jars, vibrations and such other conditions as are incident to shipping, storage, installation, and service use.

3.5.4 Maintainability. Ready access to the interior parts, terminals, and wiring of the amplifier and indicator shall be provided to permit adjustment, circuit checking, calibration, and removal and replacement of parts. If hermetic sealing is utilized, this requirement shall apply after removal of the unsealed case. The maintainability requirements shall be in accordance with MIL-M-26512.

3.5.5 Mean-time-between-failures (MTBF). The equipment shall have a MTBF of at least 346 hours with a 90 percent confidence factor (see table in MIL-R-26667 entitled "Reliability Accept-Reject Criteria (Method 2)"). Failure is any condition that results in out-of-tolerance performance.

3.5.6 Compatibility and signal interchangeability. The AVK-16/A24G-10 Indicator input requirements and ASK-10/A24G-10 Amplifier input and output requirements shall be interchangeable and compatible with the input and output characteristics of the following:

a. Computer, Central Air Data CPU-43/A in accordance with MIL-C-38037

b. AVK-16/A24G-10 Indicator manufactured by any prior alternate source

c. ASK-10/A24G-10 Amplifier manufactured by any prior alternate source

d. Transmitter, Synchro, Angle of Attack or Side Slip in accordance with MIL-T-25627.

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* 3.6 Performance. The amplifier-indicator group shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Temperature - Temperatures ranging from -55° to +71°C

b. Scale error - Scale errors and tolerances specified in tables I, II, III, and IV

c. Insulation leakage - Withstand a potential of 250V dc applied between any electrical pin contact and any metal part of the indicator case

d. Vibration

(1) Vibration error - Subjected to vibration at an amplitude of 0.009 to 0.011 inch at a frequency of 5 to 50 cps

(2) Vibration failure - Subjected to vibration at an amplitude of 0.018 to 0.020 inch at a frequency of 5 to 50 cps for a period of 3 hours

e. Temperature-altitude - Amplifier operation in accordance with figure 5

f. Slew speed - Slew speed tolerances as specified in table III

g. Altitude - Pressures ranging from 31.0 to 2.118 inches Hg

h. Acceleration - Acceleration forces up to 10g

i. Shock - Shock forces up to 15g

j. Humidity - Relative humidity up to 100 percent

k. Fungus - Exposure to simulated fungus growth as encountered in tropical climates

l. Salt spray - Exposure to simulated salt sea atmosphere

m. Sand and dust - Exposure to sand and dust particles as encountered in desert areas

n. Dielectric strength - Subjection to 250V 60 cps for 5 seconds.

3.6.1 Magnetic effect. The amplifier-indicator group shall not cause a free magnet to deflect more than 5° when the unit is held in various positions on an east-west line with its nearest part 12 inches from the center of the magnet in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted.

3.6.2 Radio noise interference. The radio noise interference requirements shall be in accordance with MIL-I-6181.

3.6.3 Insulation breakdown. The amplifier shall withstand application of 500V for 5 seconds under the conditions specified in 4.6.10. The current leakage shall be not more than 4.5 ma.

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TABLE I

Mach Section

Aircraft Mach	Transmitter Coarse°	Angle Fine°
0.2	0	0
0.4	40	200
0.6	80	40
0.8	120	240
1.0	160	80

TABLE II

Indicated Airspeed

IAS Knots	Transmitter Coarse°	Angle Fine°
80	353.33	326.67
100	0	0
200	33.3	166.67
300	66.7	333.33
400	100	140
500	133.3	306.67
600	166.7	113.33

* TABLE III

Slew Speed

Function	Maximum Time In Seconds
1. Aircraft mach tape (0.2 to 1.0)	7
2. Command mach, manual mode (with maximum switch displacement (0.2 to 1.0)) (room test only)	9
3. Indicated airspeed (50 to 600)	12
4. Command indicator airspeed manual mode (with maximum switch displacement (50 to 600))	12
5. Safe speed (angle of attack (-10 to +25))	5

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* TABLE IV

PIN CONNECTION TABLE FOR MACH-AIRSPEED INDICATOR
INDICATOR-AMPLIFIER INTERCONNECTIONS

Connector: PT02A-20-41PW PT06A-20-41SW Pin-Amplifier	Connector: PT1H-20-41PW PT06A-20-41SW To Indicator	Function	Section
W	W	Power ground	
A	A	Spare	
B	B	Spare	
V	V	Spare	
j	j	Spare	
X	X	Spare	
Y	Y	Spare	
C	C	R G out	
U	U	CT H R G out	
i	i	R G & Motor exc	SAFE SPEED
k	k	Amp out	
Z	Z	Capacitor	
D	D	5X CT H R G out	
T	T	Amp out	
s	s	Capacitor	I. A. S.
m	m	1X CT H	
E	E	R G exc	
S	S	Motor fix Ø	
h	h	Amp out	
t	t	Capacitor	
a	a	R G exc	COMM
F	F	Slew switch out	A. S.
r	r	Motor Fix Ø	
n	n	Amp Out	
R	R	Capacitor	
g	g	Motor fixed Ø	COMM.
b	b	Rate gen exc	MACH
G	G	Signal ground	
q	q	Slew switch out	
p	p	Spare	
P	P	Spare	
f	f	Spare	
c	c	1X CT H	
H	H	5X CT H R G out	
N	N	Spare	
e	e	Spare	
d	d	Amp out	MACH
J	J	Capacitor	
M	M	Motor fix Ø	
L	L	R G exc	
K	K	Chassis Ground	

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TABLE IV (Cont)

INPUTS TO INDICATOR

Connector PTIH-18-32P PT06A-18-32S Pin-Indicator	Function	Section
T A S	CT X CT Y CT Z	MACH MONITOR
U B R	CT X CT Y CT Z	SAFE SPEED
e K V	CT X CT Y CT Z	COARSE MACH
C P d	CT X CT Y CT Z	FINE MACH
f W D	CT X CT Y CT Z	COARSE IAS
N c j	CT X CT Y CT Z	FINE IAS
g X E M b h Y F	Spare CADC CADC Spare Spare Spare Lights Lights	IAS MONITOR SIGNAL (28V DC) MACH MONITOR SIGNAL (28V DC)
L a Z	CT X CT Y CT Z	IAS MONITOR
G J H	Self-test Flag trip level voltage Flag trip level voltage	(NORMALLY GROUNDED-CONNECT TO 28V DC FOR SELF-TEST) IAS MONITOR MACH MONITOR

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TABLE IV (Cont)

AMPLIFIER INPUTS AND OUTPUTS

Connector PT02A-12-10PW PT06A-12-10SW Pin Amplifier	Function
H	115V 400 cycles (H) from AC bus (/0° ref)
A	AC Common Cold (C)
B	Spare
G	Spare
K	Spare
J	Spare
C	18 ± 2 - To Angle of Attack Sensor
F	Spare
E	Spare
D	Spare

* 3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.8 Operation

3.8.1 Aircraft mach number. The aircraft mach number moving scale shall be actuated by a servo mechanism with dual speed transmitting synchro input. The input to the instrument control transformer synchro shall be linearly proportional to mach number over the range of 0.2 to 1.0. The tape scale factor shall be 1.350 inches per 0.1 mach number.

3.8.1.1 Mach number mechanization. The servo-drive mechanism shall be so designed as to accurately position the moving scale at the center reference line. Provision shall be made to prevent the moving scale from being driven beyond specified limits. When the input signal is below 0.2 mach, the movable scale shall remain fixed at 0.2 mach. Electrical zero of the followup servo-synchro shall be at mach 0.2. Electrical scale factor shall be 1,000° per unit mach for the fine synchro and 200° per unit mach for the coarse synchro. Servo slew speed shall be at least 7 mach per minute.

3.8.2 Command mach number. The command mach number symbol and associated numerical counter shall be actuated by the slew switch located below the mach tape.

3.8.2.1 Command mach number mechanization. The mechanism shall be designed to accurately position the command mach index at the appropriate position, with respect to the aircraft mach number moving scale, and this position shall also be

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displayed on the command mach counter. When the commanded mach number is beyond the visible range of the instrument dial, provision shall be made to disengage the drive mechanism; the index shall assume a position at the scale extremity and remain visible in the area marked B on figure 1. When the command index is positioned to any value on the aircraft mach number moving scale, there shall be no apparent relative motion between command index and moving scale resulting from servoed motion of the moving tape in either direction. Motion of either servoed element shall not result in interaction with the other. The reading of the command mach marker against the mach tape (when the command value is within the visible range of the mach scale) shall, at all times, read within 0.005 mach of the command value as read on the counter. When the switch is released, the servo shall provide suitable braking action to prevent coasting of the command index.

3.8.2.2 Maximum safe mach number. A fixed mach limit symbol shall be painted on the mach tape to indicate a mach limit of 0.825. The configuration and color shall be as shown on figure 1.

3.8.2.3 Indicated airspeed. The indicated airspeed moving scale shall be actuated by a servo mechanism with dual speed transmitting synchro input. The input to the synchro control transformer shall be linearly proportional to indicate airspeed over the range of 50 to 600 knots. Synchro electrical zero shall be at 100 knots. The tape scale factor shall be 1-1/4 inches per 50 knots, and the tape graduations shall be in increments of 10 knots starting at 50 knots. Numbered graduations shall be at 50-knot intervals, starting at 50 knots, with each 20-knot interval between 100 and 200 knots numbered as shown on figure 1.

3.8.2.3.1 Mechanization shall be as specified in 3.8.2.1. When the signal input is below 50 knots, the moving scale shall remain fixed at 50 knots. Electrical scale factor shall be 166.67° per 100 knots fine and 33.33° per 100 knots coarse. Servo slew speed shall be at least 2,000 knots per minute.

* 3.8.2.3.2 Maximum safe airspeed. A fixed symbol shall be painted on the tape to indicate an airspeed limit of 350 knots. The configuration and color shall be as shown on figure 1.

3.8.2.4 Command airspeed. The command airspeed index and associated numerical counter shall be actuated by the slew switch located below the IAS tape.

3.8.2.4.1 Command airspeed mechanization. The mechanism shall be designed to accurately position the command airspeed index at the appropriate position with respect to the airspeed moving scale; this position shall also be displayed on the command airspeed counter. When the command airspeed is beyond the visible range on the instrument dial, provision shall be made to disengage the drive mechanism and the index shall assume a position at the scale extremity and remain visible in the area marked B on figure 1. A center-lock position shall be provided on the command airspeed slewing switch which will position and retain the index on the center-reading reference line. In this position, the counter shall continuously provide numerical indication of indicated airspeed. The reading of the command airspeed marker against the airspeed tape (when the command value is within the visible range of the airspeed scale) shall, at all times, read within 3 knots of the command value as read on the counter. The servo shall provide suitable braking action to prevent coasting of the command index upon release of the switch.

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*3.8.2.5 Safe speed. The safe-speed moving scale shall be actuated by a servo mechanism with synchro input. The input to the indicator synchro control transformer shall represent a transmitting synchro angle range of +9° to -86°. Scale factor shall be 2.7° per 5/16 inch of tape.

*3.8.2.5.1 Mechanization shall be as specified in 3.8.2.1 except that the servo mechanism shall position the moving scale in response to the angle of attack input from a remote computer. Electrical zero shall be at 0° angle of attack. Electrical zero of the indicator followup synchro shall position the tape display to the midpoint of the stall warning band.

3.8.2.5.2 Damping. A damping adjustment shall be provided, accessible externally on the amplifier, for the purpose of adjusting to optimum damping characteristics.

*3.8.2.6 Counters. The command airspeed and mach counters shall be of the continuous-rotation type for the counter wheel displaying only the least significant digit. Counter wheels that operate intermittently shall not be used.

3.8.2.7 Slewing switches. Manually operated slewing switches as shown on figure 1 shall be provided to operate the command symbols and associated counters. The switch shall be spring centered. The slewing speed imparted to the command symbol shall be proportional to the amount the switch is displaced from center within the minimum and maximum speeds specified. The up position of the switch shall drive the command symbol and corresponding counter display toward increasing value of the function. The slew switch shall incorporate a detent that will permit indexing of one unit on the associated counters. The slew switch dimensions shall be in accordance with figure 15.

3.8.2.8 Elapsed time indicator. A 10,000-hour 4-digit type elapsed time indicator conforming to MS17322 shall be provided and so mounted in each indicator that the dial is visible on the side or rear of the case.

3.8.2.9 Decompression. The indicator shall be so designed as to withstand explosive decompression of the surrounding atmosphere. Operation shall not be affected when a pressure difference of 16 inches of mercury is applied instantaneously between the interior and exterior of the case.

3.8.2.10 Self-test circuit. A self-test circuit shall be provided in the indicator so that the mach and airspeed tapes may be positioned at a present value.

*3.8.2.11 Monitor circuits. Monitor circuits shall be provided that will detect malfunctions in the airspeed and mach components and in the intercabling, including the transmitting synchro within the air data system. Malfunction shall cause the IAS and mach warning flags to appear. A calibrated series voltage (stick-off) shall be included in the monitor circuits to provide self-test capability. This series voltage, which is equivalent to the maximum trip level, for airspeed only, shall cause the warning flag to appear when the excitation voltage is removed from the air data transmitter.

3.8.2.12 External pressure. The indicator shall withstand an external pressure of 56 inches Hg absolute without failure of the hermetic seal or cover glass, and without such distortion of the case that would affect subsequent normal operation.

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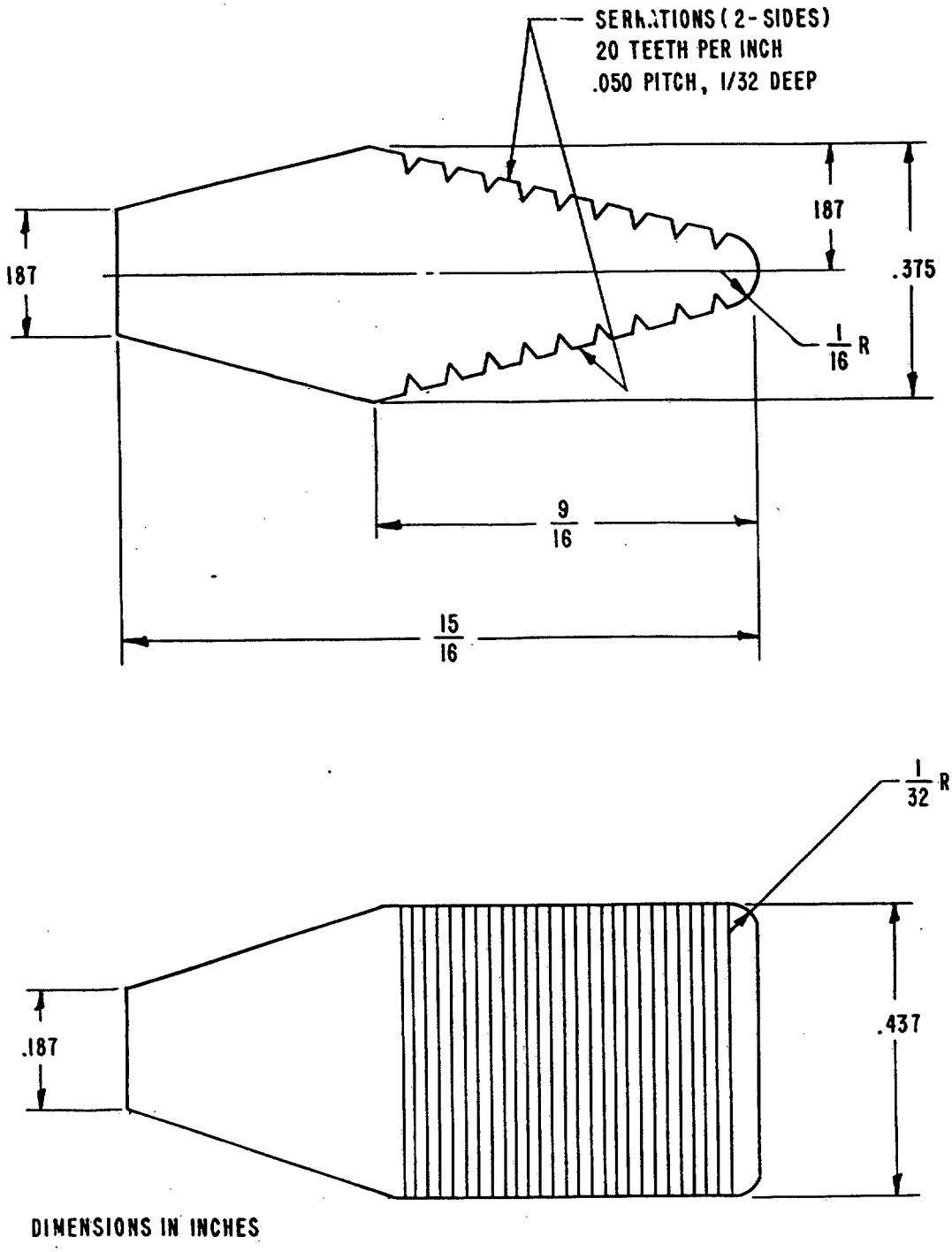


FIGURE 15. Slew Switch Knob

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3.8.2.13 Circuit isolation. The angle of attack display and transmitter circuits shall be electrically isolated, except for ground faults, from the remainder of the indicator display circuits through the use of a 1-amp fuse housed in the amplifier in an area that is not hermetically sealed.

* 3.8.3 Amplifiers. The servo amplifiers shall be transistor-mag or transistor type amplifiers. The design shall provide for plug-in modules for the power supply, auxiliary circuits, and amplifier circuits. Envelope dimensions shall not exceed those specified on figure 8. The amplifier shall be equipped with a mounting base in accordance with figure 4.

3.8.3.1 Servo-response. Response of the airspeed, mach, safe speed, and angle of attach servos shall be as follows:

Function	Bandwidth	Damping Ratio
Airspeed	3-8 cps	0.2 to 0.7
Mach	3-9 cps	0.4 to 0.7
Safe speed (Angle of attack)	Variable 2 to 5 cps	0.4 to 0.5 minimum range

* 3.9 Power requirements. The equipment shall meet all applicable requirements of MIL-STD-704 and ARINC 545 and shall operate from the power defined therein. The power requirements shall be of the following type and shall not exceed the indicated amounts:

A-C Power

Operating limits: 102 to 124V
Standby voltage limits or emergency supply: 90 to 130V
Operating frequency limits: 380 to 420 cps
Standby frequency limits or emergency supply: 380 to 420 cps
Max operating voltamperes: 98
Max warmup time: 1 minute if temperature has been at -35°C or higher, 3 minutes if temperature has been -55°C.

3.10 Lighting. The indicator shall be lighted internally in accordance with MIL-L-27160. Lighting acceptability shall be based on visual comparison with a prime standard indicator only.

3.10.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas, covering the entire face of the indicator, shall be measured for brightness level in foot-lamberts. The reference instrument and its recorded overall brightness measurements shall be approved by the procuring activity.

3.10.2 Integral lighting supply. A 400 cps voltage, variable from 0 to 5V rms, shall supply power for internal lighting for both normal and emergency operation.

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The frequency tolerances of the supply voltage shall be as specified in MIL-STD-704.

3.11 Reflective surfaces. All reflecting glass surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except that the following reflectance tolerances shall apply:

Angle of Incidence	Wave Length Millimicrons	Percent Reflectance
0°	450 to 675	0.6 Maximum
0°	425 to 700	0.5 Average
30°	450 to 625	1.0 Maximum
30°	425 to 700	0.5 Average.

3.12 Connectors. Electrical connectors used on the indicator case shall be of the hermetic-seal type and shall conform electrically to MIL-C-5015. Miniature connectors, Scintilla type as specified herein, or the electrical equivalent, shall be utilized. Keying provisions shall be made where necessary. Connector pin arrangements shall be as specified in table IV.

3.12.1 Extra contacts in receptacles. Extra contacts shall not be required unless necessary for conducting tests.

3.13 Excitation. Unless otherwise specified, excitation for all remote transmitting synchros or potentiometers shall be furnished from the amplifier-indicator.

3.14 Electron tubes. Electron tubes shall not be used.

3.15 Electronic components. The electronic components shall be in accordance with MIL-E-5400.

3.16 Case. The amplifier-indicator case shall be of nonferrous, low-density material and shall conform to figures 3, 6, and 8 with respect to dimensions and tolerances. The case shall be finished in durable dull black, color No. 37038 of FED-STD-595.

3.16.1 Case hermetic seal. The case shall provide a hermetically sealed enclosure for the internal mechanism. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. Tear strips, provided for this purpose, and filling tubes shall not have end pieces protruding outside the case dimensions. Filling tubes, if used, shall not be left exposed to possible damage from impact or tampering.

3.16.2 Filling medium. The filling medium shall be at least 98 percent pure, free of dust particles, and contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 90 percent nitrogen and the remainder helium. The absolute pressure of the filling medium shall be approximately 1/2 to 1 atmosphere as required. Where practicable, the 100 percent helium shall be utilized.

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3.17 Dial

3.17.1 Dial layout. The dial layout and dimensions shall be in accordance with figures 1 and 2.

3.17.2 Dial markings. All markings shall be so durable as to withstand usage encountered in service. The style and proportion of numerals and graduations shall conform to figure 1.

3.17.3 Numerals. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each shall be so placed that the center of mass of the numeral will be on a line joining the appropriate graduation. There shall be no doubt as to which graduation the numeral applies. When several numerals are used in one group, the space between shall be approximately 1/64 inch.

3.17.4 Dial to lighting wedge distance. The distance from dial to rear surface of lighting wedge shall not exceed 0.200 inch.

* 3.18 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.19 Weight. The weight of the indicator shall not exceed 10 pounds 8 ounces and the weight of the amplifier, including the mounting base, shall not exceed 9 pounds 2 ounces.

3.20 Finishes and protective coatings

3.20.1 Plating. Plating shall not be required on normally lubricated parts such as gears, bearings, and shafts (fabricated from brass), bronze, or corrosion-resistant steel.

3.20.2 Cadmium-plated parts. Cadmium-plated parts shall be in accordance with QQ-P-416, of a type and class to be determined by their application.

3.20.3 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film in accordance with MIL-A-8625, except that small holes, straight threads, and case inserts need not be anodized.

3.21 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The component nomenclature shall be as follows (as applicable):

Indicator, Airspeed Mach Safe Speed AVK-16/A24G-10
or
Amplifier, Electronic Control ASK-10/A24G-10.

3.22 Government loaned equipment. When specified (see 6.2), one each of the following equipment will be loaned by the Government to the contractor upon his request:

a. Computer, Central Air Data CPU-43/A in accordance with MIL-C-38037

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b. AVK-16/A24G-10 Indicator

c. ASK-10/A24G-10 Amplifier.

3.23 Workmanship. The amplifier-indicator group, including all parts and accessories, shall be sufficiently high grade to insure satisfactory operation and service life. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.23.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the equipment, they shall be held or limited accordingly.

3.23.2 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.23.3 Cleaning. After final assembly, the equipment shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, resin flash that may crumble, and other foreign material removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the amplifier and the indicator shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.2.1 Where specified, the indicator and the amplifier shall be tested as individual components.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure, temperature, and humidity are not specified, it is understood that the test will be made at atmospheric pressure (approximately 29.92 inches Hg), at room temperature (approximately 25°C), and at a maximum of 50 percent relative humidity. When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

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4.3.2 Vibration. Unless otherwise specified, the indicator shall not be subjected to any form of vibration or tapping during the performance of any of the tests specified in this specification.

4.3.3 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position.

4.3.4 Electrical power. Unless otherwise specified, the indicator shall be tested with 115V 400 cycle a-c power. The voltage and frequency shall be maintained within ±2 percent and 28V ±2 percent d-c power.

4.3.5 Standard test equipment. The indicator shall be tested with standard AY-201S-1-B synchros (or equivalent) and precision potentiometers with a resistance of 10K ohms ±5 percent, 2.5w at 25°C, and a resolution of 0.01 percent. The monitor transmitter shall be an AY50□-42-A1 or electrical equivalent.

4.3.6 Test synchros. The synchros transmitters used to provide indicator input functions shall be accurate to within ±0.1° either by selection or calibration.

4.3.7 Test potentiometers. The potentiometers used to provide indicator input functions shall have a linearity of at least 0.01 percent.

4.3.8 Standard test amplifier. The standard test amplifier shall be a standard production amplifier which has passed all individual tests specified herein.

4.3.9 Electrical zero and rotation reference. The indicator synchros shall be so oriented electrically that each function will be servoed to the indication listed in table V when a standard test transmitter is set at electrical zero and connected to the indicator and amplifier in accordance with table IV.

TABLE V

ELECTRICAL ZERO

Function	Indicator Reading With Transmitter Set at E.Z.	Standard Test Transmitter Pin Connections				
		H	C	X	Y	Z
Mach (fine synchro)	0.2 mach	26V 400 ∠4°	AC cold w/41 PW	C/32P	P/32P	a/32P
IAS (fine synchro)	100 knots	26V 400 ∠4°	AC cold w/41 PW	N/32P	c/32P	j/32P
Angle of attack	0° α	26V 400 ∠4°	AC cold w/41 PW	U/32P	B/32P	R/32P

NOTE: The ∠4° phase angle shown for the 26V 400 ∿ supply is to indicate that it is of the same phase as the 115V 400 ∿ ∠4° supply (see table IV).

4.3.9.1 The aircraft mach and indicated airspeed coarse synchros shall be checked simultaneously with their respective fine synchros. After the function is servoed to

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the EZ point by the fine autosyn, the coarse synchros shall be excited as shown on figure 9. The vacuum tube voltmeter shall indicate less than 250 mv.

4.3.9.2 A clockwise rotation of the standard test transmitter rotor (viewed from the shaft end) shall cause increasing function.

4.3.9.3 A standard test transmitter is as follows: The rotor leads are arbitrarily labeled H and C. One stator is arbitrarily labeled Z. Z is connected to C and 26V 400 cps is applied across H - C. At EZ, there will be a null voltage measured across the two unlabeled stator leads (V_1). The voltage between H and either of the unlabeled stator leads shall be 38V (approximate) (V_2). If this voltage is 15V (approximate), the synchro is 180° away from EZ (see figure 10). If clockwise rotation (viewed from the shaft end) of the synchro rotor causes V_3 to increase (within 30° of EZ), V_2 is between X and Z.

4.3.10 Potentiometer zero. Unless otherwise specified, the low end of the potentiometer shall correspond to the low end of the function.

4.3.11 Amplifier test equipment. The test equipment shall not load or affect normal operation of the unit under test. The tolerances specified are acceptable limits of performance and are not to be confused with meter accuracies. The following instruments, or their equivalents, may be used:

a. Vacuum Tube Voltmeter Model 400C
Hewlett-Packard Co.
Palo Alto, California

b. Megger
James Biddle Co.
Philadelphia, Pa.

c. Cathode-Ray Oscillograph Type 304A
Allen B. Dumount Laboratories, Inc.
Clifton, N. J.

d. Voltohmmeter-Senior Volt Ohmyst WV-98A
Radio Corporation of America
Harrison, N. J.

e. Phase Comparator.

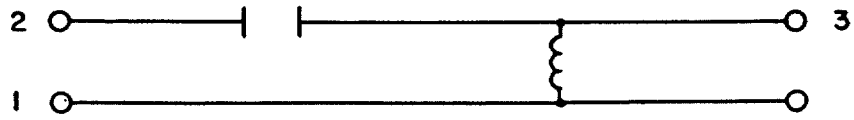
4.3.11.1 Amplifier attitude. Unless otherwise specified, the unit under investigation shall be tested in its normal operating position.

4.3.11.2 Power supply

4.3.11.2.1 Alternating current. The amplifier shall be tested with 115 ±3V rms, 400 cycles, single phase, with a total harmonic distortion less than 5 percent and a maximum frequency deviation of less than 1 percent, capable of supplying 150 va.

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4.3.11.2.2 Phase. A voltage specified as being "from pin X to pin Y" shall be measured, using pin X as a reference or ground terminal. The direction of leading phase angle can be established by use of the following circuit:



In this circuit, the voltage from pin 1 to pin 3, V_{13} leads the voltages from pin 1 to pin 2, V_{12}. All phase measurements shall be taken with respect to the a-c excitation measured from pin A to pin H on connector PT02A-12-10PW.

4.3.11.2.3 Direct current. Twenty-eight ±1V capable of supplying 250 ma shall be applied. Ripple and noise on direct current shall not exceed 5 percent.

4.3.11.3 Signal source (AC). The 400-cps signal source shall have the following characteristics:

a. An output impedance not exceeding 1,000 ohms

b. A voltage range of 0 to 1V continuously variable

c. A phase of 0° and 180° ±3°, selectable

d. A distortion not exceeding 5 percent.

4.3.11.4 Signal source (DC). The d-c signal source shall have the following characteristics:

a. An output resistance not exceeding 1,000 ohms

b. A voltage range of 0 to 1V continuously variable

c. Polarity - Both polarities shall be available (positive and negative)

d. Ripple - The total rms voltage appearing at the signal source shall not exceed 2.0 mv.

4.3.11.5 Loads. Unless otherwise specified, the loads (Z) specified herein shall consist of 155 $\angle 40°$ ±1 percent ohms, measured at 26V 400 cps unless otherwise specified.

4.4 Preproduction testing

4.4.1 Test samples. The preproduction test samples shall consist of three amplifiers and three indicators representative of the preproduction equipment. Samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The samples that were tested shall be submitted to the procuring activity.

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4.4.2 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 Individual tests (indicator). Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Functional check

c. Synchro correspondence

d. Indicator case leak

e. Dielectric strength

f. Scale error

g. Warning flag

h. Slew speed

i. Lighting.

4.5.1.1 Individual tests (amplifier). Each amplifier shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Insulation breakdown

c. Leakage test

d. Circuit connections

e. Voltage and current measurements

f. Null

g. Gain

h. Phase shift

i. Saturation

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j. Power consumption

k. Functional test.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A (indicator). One indicator, selected at random from each 100 or less produced on the contractor or order, shall be subjected to the following tests as described under 4.6:

a. Individual tests

b. Low temperature operation, indicator

c. High temperature operation, indicator

d. Vibration error

e. Frequency response test

f. Lighting.

4.5.2.1.1 Sampling plan A (amplifier). One amplifier, selected at random from each 100 or less produced on the contractor or order, shall be subjected to the following tests as described under 4.6:

a. Individual tests

b. Low temperature operation, amplifier

c. High temperature operation, amplifier

d. Vibration.

* 4.5.2.2 Sampling plan B (indicator). Unless otherwise specified, 3 indicators shall be selected at random from the first 15 items on the contract or order and subjected to the following tests as described under 4.6. When a change is made in design or material which affects the performance of the indicator in any of these tests, the tests so affected shall be performed on the number of new units specified above.

a. Sampling plan A tests

b. Magnetic effect

c. Vibration failure, indicator

d. Acceleration, indicator

e. Humidity, indicator

f. Fungus resistance, indicator

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g. Salt spray, indicator

h. Altitude

i. Sampling plan B tests of MIL-L-27160

j. Reliability

k. Decompression test

l. Compatibility and signal interchangeability tests

m. External pressure.

4.5.2.2.1 Sampling plan B (amplifier). Unless otherwise specified, 3 amplifiers shall be selected at random from the first 15 items on the contract or order and subjected to the following tests as described under 4.6. When a change is made in design or material which affects the performance of the amplifier in any of these tests, the tests so affected shall be performed on the number of new units as specified above.

a. Sampling plan A tests

b. Extreme temperature exposure

c. Temperature-altitude test

d. Vibration failure, amplifier

e. Shock

f. Acceleration, amplifier

g. Humidity, amplifier

h. Radio noise interference

i. Salt spray, amplifier

j. Fungus resistance, amplifier

k. Magnetic property, amplifier

l. Reliability

m. Compatibility and signal interchangeability tests.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

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4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator and the amplifier shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and markings.

4.6.2 Functional check. The functional check shall be performed prior to final sealing and final inspection of the indicator. The indicator functions shall be cycled simultaneously at the rate of 1 cycle every 2 minutes for a minimum of 2 hours. A cycle shall consist of the excursion from minimum value of the function to maximum value, and return to minimum value. The indicator shall meet all requirements after completion of the functional check.

4.6.3 Synchro correspondence. Any indicator section that has a dual speed input shall have these synchros inspected to assure that the coarse synchro is set in satisfactory correspondence with its associated fine synchro. The coarse synchros shall be considered to be in satisfactory correspondence when the rotor output is less than 150 mv with the stator excited as on figure 9 for a fine synchro setting of precisely EZ. The 10V supply may be replaced by a precision synchro transmitter set precisely at EZ.

4.6.4 Leak tests

4.6.4.1 Indicator case leak. The indicator shall be filled with a dust-free 10-percent-helium 90-percent-nitrogen mixture at a pressure of approximately 1 atmosphere and with a dewpoint below -65°C. Following final sealing, the indicator shall be tested for case leaks. The leak rate shall not exceed 7.5 micron cubic feet per hour.

4.6.4.2 Leakage test. Each module of the amplifier unit shall be marked indicating that it has been subjected to a leakage test and that the leakage rate of helium, as detected by a mass spectrometer, did not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.6.5 Dielectric strength. The application of 250V 60 cps between the pins and the shell of the connector for 5 seconds shall cause no evidence of voltage breakdown. Pins to be exempted from this test shall be identified in installation drawings.

4.6.6 Scale error

4.6.6.1 Aircraft mach. The coarse synchro shall be set at the value specified in table I. The tape shall be brought into the specified reading using the fine synchro only and being careful to approach from the right direction.

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4.6.6.1.1 Mach accuracy. Test readings shall be obtained from both function increasing and function decreasing. No readings shall be in error in excess of 5.0° (0.005 mach) as read on the fine synchro transmitter.

4.6.6.2 Command mach correlation. Not only shall each presentation meet its individual accuracy requirements but they shall also maintain close correlation between themselves.

4.6.6.2.1 Command mach correlation accuracy. The marker reading shall not differ from the counter reading at any test point by more than 0.005 mach.

4.6.6.2.2 Command mach disengagement. When the transfer function commands a position beyond the viewing range of the indicator, the marker shall position itself at the window extremities, in view, and resume indexing against the aircraft mach tape when that portion of the tape is stationed in the window area.

4.6.6.2.3 Command mach disengagement test. With dual speed synchros set in 0.2 mach on the mach tape, the following shall apply:

a. Set in 1.0 mach with the command slew switch. The marker shall position itself at the lower extremity of the window.

b. Set in 1.0 mach with the dual synchro inputs.

c. The tape shall so position the marker that some part of the marker is obscured by the lubber line.

d. Set in 0.2 mach with the command slew switch.

e. The marker shall position itself at the upper extremity of the window.

f. Set in 0.2 mach with the dual synchro inputs.

g. The tape shall so position the marker that some part of the marker is obscured by the lubber line.

h. The counter shall display the command mach throughout.

4.6.6.3 Manual mode. The manually operated slew switch below the mach window actuates the command marker and the command counter. The up position of the switch shall increase command function. The down position shall decrease command function. The rate of slew shall be proportional to switch displacement.

4.6.6.4 IAS (indicated airspeed). The coarse synchro shall be set at the values specified in table II. The tape shall be brought into the specified reading using the fine synchro only, being careful to approach from the right direction.

4.6.6.4.1 IAS accuracy. Test readings shall be obtained for increasing function. No readings shall be in error in excess of 1.67° (1 knot) as read on the fine synchro transmitter. The following stations shall also be checked for a decreasing function: 200, 400, and 600 knots.

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4.6.6.5 Command IAS correlation. Not only shall each presentation meet its individual accuracy requirements, but they shall also maintain close correlation between themselves.

4.6.6.5.1 Command IAS correlation accuracy. The marker reading shall not differ from the counter reading at any station by more than 3 knots.

4.6.6.5.2 Command IAS disengagement. When the transfer function commands a position beyond the viewing range of the indicator, the marker shall position itself at the window extremities, in view, and resume indexing against the tape when that portion of the tape is stationed in the window area.

4.6.6.5.3 Command IAS disengagement test. With dual speed synchros set in 100 knots on IAS tape, the following shall apply:

a. Set in 600 knots with the command slew switch.

b. The marker shall position itself at the lower extremity of the window.

c. Set in 600 knots with the dual synchro inputs.

d. The tape shall so position the marker that some part of the marker is obscured by the lubber line.

e. Set in 100 knots with the command slew switch.

f. Set in 100 knots with the dual synchro inputs.

g. The tape shall so position the marker that some part of the marker is obscured by the lubber line.

h. The counter shall display the command IAS throughout.

4.6.6.6 Manual mode. The manually operated slew switch below the IAS window actuates the command marker and command counter when it is moved in the vertical plane. The up position of the switch shall increase command function and the down position shall decrease command function. The rate of slew shall be proportional to switch displacement.

4.6.6.7 Centering transmitter. The side-throw position of the IAS switch shall detent the lock into place and actuate and position the command marker to the lubber line to within ±1 knot.

4.6.6.7.1 The centering transmitter may be checked in conjunction with the scale error test for the IAS tape.

4.6.6.7.2 Center-lock operation. With the centering transmitter engaged, the command IAS counter shall continuously provide numerical indication of the airspeed tape reading.

4.6.6.7.3 Centering accuracy. With the transmitter set at the angles listed in table II, the counter readings shall be accurate to within 3 knots, except that between

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the range of 140 knots to 250 knots, the readings shall be within ±1 knot. Readings shall be taken in an increasing direction. The following stations shall also be checked in a decreasing direction: 200, 400, and 600 knots.

*4.6.6.8 Safe speed (angle of attack). The test points listed below shall be brought under the lubber line, being careful to approach from the right direction. The transmitter angle shall be recorded.

Test Points	Transmitter Angles
	EZ
Midpoint of stall warning band	
Apex of triangle (1.2)	-35.1°
Base of triangle (1.3)	-51.3°
Base of rectangle (1.4)	-66.15°

*4.6.6.8.1 Safe speed accuracy. Readings shall be taken in both increasing and decreasing directions. No reading shall be in error in excess of a central transformer followup synchro angle equivalent to 0.75°.

4.6.6.9 Position error. There shall be no shift of any of the presentations due to indicator attitude. With all servoes at null, change the attitude of the indicator, then observe and record change in any reading.

4.6.7 Mach warning flag. When the indicator is connected to the test amplifier and the mach tape satisfies the input signal, the flag shall disappear from view.

4.6.7.1 IAS warning flag. When the indicator is connected to the test amplifier and the IAS tape satisfies the input information, the flag shall disappear from view.

4.6.7.2 Self-test circuit. Connecting pin G of PT1H-18-32P to 28 ±5V dc (by closing self-test switch) shall result in the mach tape being positioned at 0.92 ±0.005 mach number and the airspeed tape being positioned at 225 ±2.0 knots. The low side of the 28V dc shall be connected to frame ground. During the period in which the tapes are being positioned, the IAS warning flag shall be in view if the IAS is at least 4 knots from the self-test point at initiation of the test. The mach warning flag shall be in view if the mach function is at least 0.005 mach from the self-test point.

4.6.7.3 Monitoring circuit. The airspeed and mach monitor shall detect the following malfunctions:

a. Inaccuracies in excess of 4 knots and 0.007 mach
(Note: The monitor shall not trip for inaccuracies less than 2 knots or 0.003 mach and shall not produce nuisance trips during the functional test (4.6.18))

b. Fine transmitter or C. T. stator or rotor failure due to actual component failure or intercabling failure

c. Loss of power to fine transmitter

d. Coarse transmitter or C. T. stator failure due to actual component failure or intercabling failure

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e. Slippage of the airspeed or mach tape resulting in an error of 4 knots or 0.007 mach

f. Removal of excitation power to the monitor transmitter for airspeed only.

Rotation of the monitoring test transmitter to angles listed in tables I and II (fine transmitters) 3.33° shall not cause the monitor flags to appear. Rotation of the test transmitters to angles of 6.67° from those listed in the tables shall cause the warning flags to appear. This test shall be performed at the 100-, 300-, and 500-knot test values and at the 0.4-, 0.6-, and 0.8-mach test values.

4.6.8 Slew speed. Using a stop watch, record the time for full excursion of the various presentations listed in table III. The times shall not exceed those specified for either increasing or decreasing function.

4.6.9 Lighting

4.6.9.1 Visual comparison. Indicator lighting shall be checked by comparison with a prime standard as defined in 3.10.1. The prime standard and the instrument under test shall be placed side by side, at eye level. Both units shall be connected to a common variable source (0-5V). The brightness of the two units shall be compared with 2.85V ±0.1V applied to the standard and 2.85V ±0.1V applied to the unit under test. If the brightness lighting of the two units appears the same visually, the unit under test shall be considered acceptable. If any area of the instrument under test appears visually to differ from the same area of the prime standard, it shall then be subjected to all display tests listed in MIL-L-27160 and sufficient data shall be taken to substantiate conformance to the display requirements of MIL-L-27160.

4.6.9.1.1 Color comparison. Apply 4.5V to the prime standard and the indicator being tested. The two indicators shall be compared and shall visually appear identical in color.

4.6.9.2 Stray light. The lighting system shall be housed to prevent the leakage of stray light and to shield all lamp filaments from direct view. Stray light measurements shall be made as follows: A flat white reflecting surface of 85 ±5 percent reflectance shall be placed perpendicular to the front cover glass and parallel to the top edge of the mounting flange. Brightness readings shall be made looking down (perpendicular) on this reflecting surface and 1 inch in front of the cover glass. When the reflecting surface is positioned in the lower half of the display, the readings shall not exceed 1.5 foot-lamberts, and when positioned in the upper half, shall not exceed 0.2 foot-lambert. Measurements shall be repeated looking upon the reflecting surface. In no case shall the readings exceed 0.2 foot-lambert. All measurements shall be made at 4.5 ± 0.1V.

4.6.10 Insulation breakdown. This test shall be performed with the equipment de-energized and not connected to the rest of the circuitry, and with the chassis ground lugs removed from ground. Application of the specified sine wave rms voltage at commercial frequency between each terminal designated and the case for a period of 5 seconds shall not cause insulation breakdown. The current leakage shall not exceed 4.5 ma.

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Test Voltage	Connector	Pin
500V	10PW	A

CAUTION - This is a destructive test and shall not be performed repeatedly.

4.6.11 Circuit connections. The airspeed, mach, and safe speed amplifier shall consist of 5 separate signal channels (aircraft mach, command mach, indicated airspeed, safe speed, and command airspeed channels) having a common power transformer. Each channel shall be tested separately. The load on each channel shall be 155 ohms ∠40° ±1 percent. The common ground shall be used for all loads. Connection between output pins shall be removed when the load is removed.

4.6.12 Voltage and current measurements

* 4.6.12.1 The a-c voltages and their corresponding phase angles shall be measured with respect to reference phase from ground to each of the following designated pins:

Connector	Pin	RMS Voltage	Phase
41PW	S	26 ±2 percent	0° ±5°
10PW	C	18 ±2V	0° ±5°
41PW	i	18 ±2V	0° ±5°
41PW	b	9 ±2 percent	0° ±5°
41PW	M	26 ±2 percent	0° ±5°
41PW	L	9 ±2 percent	0° ±5°
41PW	g	26 ±2 percent	0° ±5°
41PW	E	9 ±2 percent	0° ±5°
41PW	a	9 ±2 percent	0° ±5°
41PW	r	26 ±2 percent	0° ±5°

4.6.12.2 IAS and mach flag circuits. With proper application of power to the system, the flags shall disappear from view.

4.6.13 Null

4.6.13.1 Safe speed, amplifier. With the input shorted, the total rms voltage across the load shall not exceed 900 mv.

4.6.13.2 Command mach, amplifier. With the a-c input shorted, the total rms voltage across the load shall not exceed 900 mv.

4.6.13.3 Aircraft mach, amplifier. With the fine input shorted, the total rms voltage across the load shall not exceed 900 mv.

4.6.13.4 Indicated airspeed amplifier. With the fine input shorted, the total rms voltage across the load shall not exceed 900 mv.

4.6.13.5 Command airspeed, amplifier. With the a-c input shorted, the total rms voltage across the load shall not exceed 900 mv.

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4.6.14 Gain

4.6.14.1 Safe speed, amplifier. With an input signal of 10 mv (0° and 180°), the output shall be 5 ±1.5V.

4.6.14.2 Command mach, amplifier. With an input signal of 10 mv (0° and 180°), the output shall be 5 ±1.5V.

4.6.14.3 Aircraft mach, amplifier. The gain of the amplifier with the signal applied to the coarse input terminals and fine input terminals shall be as follows:

a. These tests shall be made for both polarities

b. Fine input - The output shall be 5 ±1.5V when 12 mv is applied to the fine input terminals

c. Coarse input - When the coarse input is 150 mv, the output shall not exceed 1.5V. When the coarse input is 1.0V, the output shall be 5V minimum.

4.6.14.4 Command airspeed. With an input signal of 10 mv ac (0° and 180°), the output shall be 5 ±1.5V.

4.6.15 Phase shift

4.6.15.1 Safe speed, amplifier. The output voltage shall lead the input signal voltage by 90° ±15°. Tests shall be performed with a 10 mv signal voltage applied in phase with the reference excitation voltage and 180° away from this reference voltage.

4.6.15.2 Command mach, amplifier. The 400 cps output voltage shall lead the 400-cps input signal voltage by 90° ±15°. Tests shall be performed with a 10-mv input signal voltage applied in phase with the reference excitation voltage and 180° away from this excitation voltage.

4.6.15.3 Aircraft mach, amplifier. The output voltage shall lead the input signal voltage by 90° ±15°. Tests shall be performed with a 12-mv signal voltage applied in phase with the reference excitation voltage and 180° away from this reference voltage. The signal shall be applied to the fine input terminals.

4.6.15.4 Indicated airspeed, amplifier. The output voltage shall lead the input signal voltage by 90° ±15°. Tests shall be performed with a 12-mv signal voltage applied in phase with the reference excitation voltage and 180° away from this reference voltage. The signal shall be applied to the fine input terminals.

4.6.15.5 Command airspeed, amplifier. The 400-cps output voltage shall lead the 400-cps input signal voltage by 90° ±15°. Tests shall be performed with a 10-mv input signal voltage applied in phase with the reference excitation voltage and 180° away from this excitation voltage.

4.6.15.6 Indicated airspeed, amplifier. The gain of the amplifier with the signal applied to the coarse input terminals and fine input terminals shall be as follows:

a. These tests shall be made for both polarities

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b. Fine input - The output shall be 5 ±1-1/2V when 12 mv is applied to the fine input terminals

c. Coarse input - When the coarse input is 150 mv, the output shall not exceed 1.5V. When the coarse input is 1.0V, the output shall be 5V minimum.

4.6.16 Saturation

4.6.16.1 Safe speed amplifier. The output shall be not less than 13V when the input signal is 200 mv of either phase.

4.6.16.2 Command mach amplifier. The output shall be not less than 13V when the input signal is 200 mv rms of either phase.

4.6.16.3 Aircraft mach, amplifier. The output voltage shall be not less than 13V when the fine input is 200 mv of either phase.

4.6.16.4 Indicated airspeed, amplifier. The output voltage shall be not less than 13V when the fine input is 200 mv of either phase.

4.6.16.5 Command airspeed, amplifier. The output shall be not less than 13V when the input signal is 200 mv rms of either phase.

4.6.17 Power consumption. With the amplifier excited with an applied voltage of 115V 400 cps and all loads connected, the power consumption shall be not greater than 98 va.

4.6.18 Functional test. The amplifier shall be excited and all chennels shall be used to simultaneously cycle indicator functions at the rate of 1 cycle each 2 minutes for 2 hours. A cycle shall consist of the excursion from minimum value of the indicator function to maximum value and return to minimum value.

4.6.19 Low temperature operation, indicator. The indicator shall be subjected to a temperature of -54° ±2°C for a period of 4 hours with no electrical power applied. At the end of that period and after a 3-minute warmup, while still at the specified temperature, the unit shall meet the tests specified in 4.6.6.1, 4.6.6.1.1, 4.6.6.2, 4.6.6.2.1, 4.6.6.3, 4.6.6.6, 4.6.6.7.1, 4.6.6.8, 4.6.6.8.1, 4.6.7, and 4.6.8, except that the permissible tolerance shall be increased by 75 percent.

4.6.20 Low temperature operation, amplifier. The amplifier shall be subjected to a temperature of -54° ±2°C for a period of 4 hours. After exposure with the temperature maintained at -54°C and following a 3-minute warmup period, the unit shall meet the requirements of 4.6.12.1 and 4.6.13 through 4.6.17 except that the following tolerances shall apply:

Null: 1.5V maximum
Gain: AC 5 ±2V
Phase shift: 90° ±30° and 270° ±30°
Saturation: 13V minimum

*4.6.21 High temperature operation, indicator. The indicator shall be subjected to a temperature of +71° ±2°C for a period of 4 hours with no electrical power applied. At the end of this period and while still at the specified temperature, the unit

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shall meet the requirements of 4.6.6.1, 4.6.6.1.1, 4.6.6.2, 4.6.6.2.1, 4.6.6.3, 4.6.6.6, 4.6.6.7, 4.6.6.7.1, 4.6.7, and 4.6.8, except that the permissible tolerance shall be increased by 75 percent.

4.6.22 High temperature operation, amplifier. The amplifier shall be subjected to a temperature of +71° ±2°C for a period of 4 hours. The amplifier shall then meet the requirements of 4.6.12.1 and 4.6.13 through 4.6.17 except that the following tolerances shall apply:

Null: 1.5V maximum with a-c inputs shorted
Gain: AC 5 ±2V
Phase shift: 90° ±30° and 270° ±30°
Saturation: 13V minimum

4.6.23 Vibration error. While operating, the indicator shall be subjected to a vibration test in accordance with procedure IV of MIL-E-5272 at amplitudes of 0.009 to 0.011 inch. Permissible tolerances during vibration shall not exceed the room temperature scale error tolerances specified in 4.6.6. Scale error shall be determined at not less than three test points for each servo.

4.6.24 Vibration. With the amplifier operating, the test shall be conducted without shock mounts and as specified in procedure XII of MIL-E-5272, curve B. Only the cycling portion of procedure XII shall be conducted. There shall be no physical damage to the unit as a result of the test. During this test, each channel of the amplifier shall meet the tests specified in 4.6.12 through 4.6.18.

4.6.25 Frequency response test. Response tests shall be performed for each servo system within the amplifier-indicator. Figure 11 depicts a typical circuit for obtaining the necessary data. The bandwidth and damping factor shall be determined by plotting the ratio of phase-out to phase-in versus frequency as specified on figure 12. For a servo less than critically damped, the damping ratio is related to response curve peak by the equation:

$$\text{For } \delta < 1,\ Mp = \frac{1}{2\delta\sqrt{1-\delta^2}}$$

where Mp = Magnitude of peak

and δ = damping ratio

4.6.26 Magnetic effect. The magnetic effect of the unit shall be determined in terms of deflection of a free magnet approximately 1-1/2 inches long, suspended in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted. It shall not cause the free magnet to deflect more than 5° when the unit is held in various positions on an east-west line with its nearest part 12 inches from the center of the magnet. Tests shall be made with power applied to the unit.

4.6.27 Vibration failure, indicator. While operating, the unit shall be subjected to a vibration failure test in accordance with procedure V of MIL-E-5272.

4.6.28 Acceleration, indicator. While operating, the indicator shall be subjected to an acceleration test in accordance with procedure III of MIL-E-5272 at 7g.

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4.6.29 Acceleration, amplifier. The amplifier shall be subjected to the test specified in procedure III of MIL-E-5272 at 10g. During the test, the amplifier shall meet the requirements of 4.6.12.1 and 4.6.13 through 4.6.17.

4.6.30 Humidity, indicator. The indicator or a suitable hermetically sealed mockup shall be subjected to a humidity test in accordance with procedure I of MIL-E-5272 with relative humidity up to 100 percent.

4.6.31 Humidity, amplifier. The amplifier shall be subjected to the test specified in procedure I of MIL-E-5272. At the completion of the test, each amplifier shall meet the requirements of 4.6.12.1 and 4.6.13 through 4.6.17.

4.6.32 Fungus resistance, indicator. The indicator, or a suitable hermetically sealed mockup that exactly duplicates the external portion of the unit, shall be subjected to fungus growth as encountered in a tropical climate in a mold chamber maintaining an internal temperature of 30°C and a relative humidity of 95 percent for a period of 28 days. At the end of this test, the parts shall be visibly examined. Deterioration or corrosion of any internal components which could in any manner prevent the indicator meeting operational requirements during service life shall be considered failure.

4.6.33 Fungus resistance, amplifier. The amplifier shall be subjected to the fungus resistance tests as specified in procedure I of MIL-E-5272. At the completion of the tests, the amplifier shall not suffer deterioration which could in any manner prevent the equipment from meeting operational requirements of 4.6.12.1 and 4.6.13 through 4.6.17.

4.6.34 Salt spray, indicator. The indicator or a suitable hermetically sealed mockup that exactly duplicates the external portion of the indicator shall be subjected to a salt spray test in accordance with procedure I of MIL-E-5272.

4.6.35 Salt spray, amplifier. The amplifier shall be subjected to the salt spray test as specified in procedure I of MIL-E-5272. At the completion of the test, the unit shall meet the requirements of 4.6.12.1 and 4.6.13 through 4.6.17.

4.6.36 Altitude. The indicator shall be subjected to a pressure of 2.118 inches Hg. At this pressure, all functions of the indicator shall be capable of operating through full scale. After being returned to sea level pressure, the indicator shall be capable of meeting the requirements of 4.5.1.

4.6.37 Extreme temperature exposure. After alternate exposures to temperatures of -54°C and +71°C for periods of 24 hours each and a delay of 3 hours at room temperature following completion of this exposure, the amplifier shall meet the requirements of 4.6.12.1 and 4.6.13 through 4.6.17.

4.6.38 Temperature-altitude tests. The temperature-altitude tests shall be conducted in accordance with the conditions of figure 5. At any combination of temperature-altitude specified on curves A, B, and C, the amplifier shall meet the

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requirements of 4.6.11, 4.6.12.1, and 4.6.13 through 4.6.17 except that the following tolerances shall apply:

Null: 1.5V maximum with a-c input shorted
Gain: AC 5 ±2V
Phase shift: 90° ±30° and 270° ±30°
Saturation: 13V minimum

4.6.39 Shock. The amplifier shall be subjected to the tests as specified in procedure V of MIL-E-5272. The amplifier shall meet the requirements of 4.6.12.1, and 4.6.13 through 4.6.17. The amplifier shall suffer no mechanical failures due to the applied shocks.

4.6.40 Radio noise interference. The amplifier shall be subjected to the requirements of MIL-I-6181 except that the upper frequency limit of the radiated noise requirements shall be 1,000 mc per second. The conducted noise requirements shall not apply to the frequency range nor shall they apply to those conductors that interconnect components.

4.6.41 Magnetic effect. The amplifier shall be magnetically shielded. When held in any position so that the nearest portion of the amplifier is a distance of 12 inches from the pivot of the card of an aircraft magnetic compass in a horizontal magnetic field having a strength of 0.18 oersted, the amplifier shall not cause a change in reading of the compass card to exceed 5°.

4.6.42 Reliability. A minimum of three amplifiers and three indicators shall be subjected to the following reliability test. The maximum allowable failures for the test time specified shall be in accordance with table VI.

TABLE VI

Maximum Allowable Failures	Total Test Time (3 Units)
2	2,076
3	2,352
4	2,629
5	2,941

Each indicator and amplifier shall be tested a minimum of 693 hours of actual electrical operation with a cycling time for each indicator function of 6 to 7 minutes per cycle with a minimum of 6,000 cycles per function. A cycle shall consist of the excursion from minimum value of the function to maximum and return to minimum. The test shall begin when rated power is applied to the unit. After 4 hours of rated power on, the power shall be turned off for a period of 2 hours. This procedure of power-on, power-off operation shall be repeated until the test has been completed. Only the power-on portion of the test shall be utilized in calculating the total test time specified in table VI. After each 50 hours of electrical operation, the system shall be subjected to a scale error test of at least three test stations per function. Room tolerances shall apply throughout. The indicator shall be tested at room temperature for approximately 64 percent of the test time, at -10°C for approximately 14 percent of the test time, and at 50°C for approximately 22 percent of the test time as shown in table VII.

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TABLE VII

Reliability Test Table

Cycling Period	Number of Hours	Ambient Temp (°C)
1	50	25
2	50	50
3	100	25
4	50	-10
5	50	25
6	50	50
7	100	25
8	50	-10
9	100	25
10	50	50
11	50	25

During the fifth cycle, the indicator shall be vibrated in accordance with procedure IV of MIL-E-5272 at an amplitude of 0.009 to 0.011 inch and a frequency of 35 cps. The transition from the fourth to the fifth cycling period shall be completed in the shortest possible time to allow the vibration test to be conducted during the temperature change from -10° to +25°C. In case of malfunction of any kind, the malfunction shall be recorded and the test interrupted while repairs are being made. The test shall then be continued from the point of interruption. Failure to meet the scale error tests shall be considered a malfunction. Upon completion of this test, the indicator shall again be tested for scale error, and the allowable tolerances increased by 50 percent. If recalibration is required to pass the scale error test, an additional malfunction shall be charged. The test may be interrupted for normal weekend shutdowns. All tests listed under 4.6 may be included as part of this test. Satisfactory completion of the reliability test (the maximum failures shall not exceed those specified in table VI) will demonstrate with a 90 percent confidence level that an MTBF of at least 346 hours has been achieved (MIL-R-26667, table entitled "Reliability Accept-Reject Criteria (Method 2)").

4.6.43 Sand and dust. The unit shall be subjected to a sand and dust test in accordance with MIL-E-5272, procedure I. At the completion of this test, the unit shall operate satisfactorily.

4.6.44 Decompression test. The indicator shall be placed in a cabinet in which the pressure is maintained the same as inside the hermetically sealed case. The pressure within the cabinet shall be released in such a manner that a pressure differential of 16 inches Hg is instantaneously applied between the interior and exterior of the case. Operation of the indicator shall not be affected by the sudden decrease in pressure.

4.6.45 External pressure. The indicator shall be subjected to a pressurization test by applying an absolute pressure of 56 inches Hg (approximately 26 inches Hg above atmospheric pressure) for a period of 15 minutes with rated power on the indicator. Normal subsequent operation shall not be affected by this test and no failure of the cover glass, hermetic seal, or case shall occur.

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4.6.47 Compatibility and signal interchangeability tests. Compatibility and signal interchangeability tests (see 6.2), as required to demonstrate compliance with 3.5.6 shall be conducted in accordance with the applicable document for the equipment specified herein.

4.6.47 Maintainability verification. Maintainability verification shall be in accordance with MIL-M-26512, appendix A, paragraph entitled "Maintenance Task Test and Demonstration". All tasks selected shall be effectively simulated.

*4.7 Inspection of preservation, packaging, packing, and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with section 5 or the documents specified therein.

*5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Each amplifier-indicator group shall be preserved in accordance with MIL-P-116, method III and packaged in accordance with MIL-P-7936.

5.1.2 Level C. The amplifier-indicator group shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.1.3 Verification. The rough handling tests of MIL-P-7936 or an engineering analysis for level A or C packaging shall verify that the unit package will limit shock forces reaching the contained amplifier-indicator group to or below 15g (see 3.6, item i).

5.2 Packing and marking

5.2.1 All levels. Amplifier-indicator groups preserved and packaged as specified in 5.1.1 or 5.1.2 shall be packed and marked in accordance with MIL-P-7936 and MIL-STD-129, respectively.

6. NOTES

6.1 Intended use. The A/A24G-10 amplifier-indicator group covered herein is intended for use within the cockpit of a high performance weapon system.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When guide pins are required (see figure 3)

c. When Government loaned equipment will be available for loan to the contractor (see 3.22)

d. When sampling plan B tests will not be conducted

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e. Procedure for conducting compatibility and signal interchangeability tests (see 4.6.46)

f. Levels of packaging and packing required (see section 5).

6.3 The outside margins of this specification have been marked to indicate where changes from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 11, 67

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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